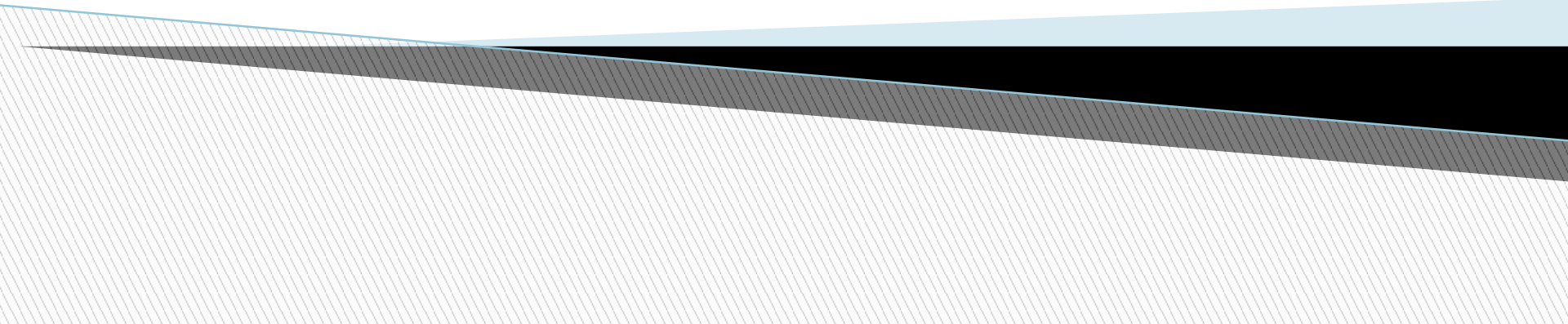


Database Design Process



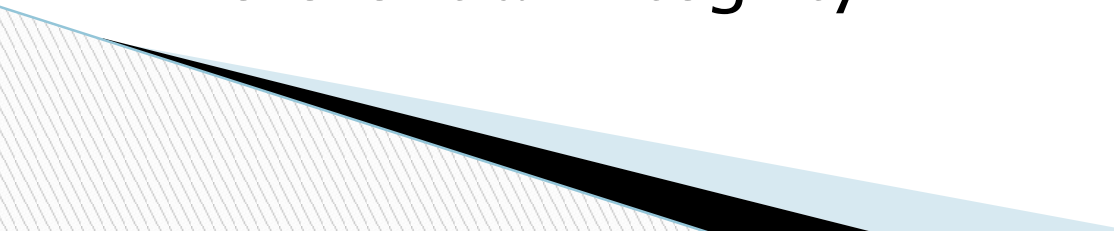
Data Modeling

Data modeling is essential to building a well-functioning database. For a database to support the activities of a business, it needs a good blueprint and foundation: the data model. A data model represents a business' data.

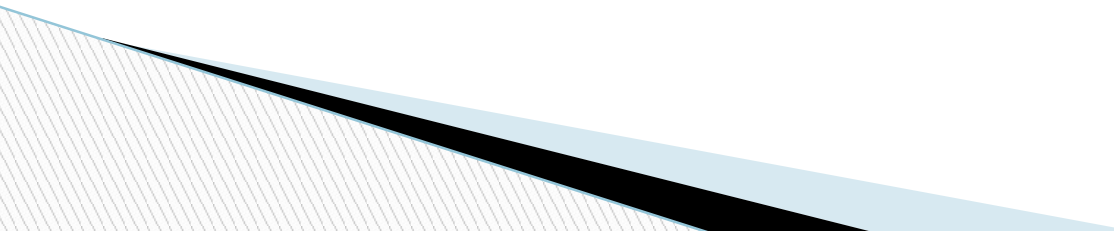
Data Modeling

Effective database designers will keep in mind the principles of normalization while they design a database.

Normalization is a database design approach that seeks the following four objectives:

- minimization of data redundancy,
 - minimization of data restructuring,
 - minimization of I/O by reduction of transaction sizes, and enforcement of referential integrity.
- 

Data Modeling

- ▣ *A poor database design can cripple an application, producing problems with redundancy, inaccuracy, consistency, and concurrency of your data.*
 - ▣ *Normalization is a process that serves to reduce, if not eliminate, these problems with data.*
- 

Database Design

The database design process can be broken down into four phases.

- Phase 1 - Requirements Collection and analysis phase
- Phase 2 - Conceptual Design
- Phase 3 - Logical Design
- Phase 4 - Physical Design

Database Design...



Phase 1 - Requirements Collection and Analysis phase

Functional
Requirements



Database
Requirements



Prospective database users are interviewed to understand and document their data requirements.

From data view (e.g. ERD) rather than functional view (e.g. DFD)



Database Design...

Phase 2 - Conceptual Design

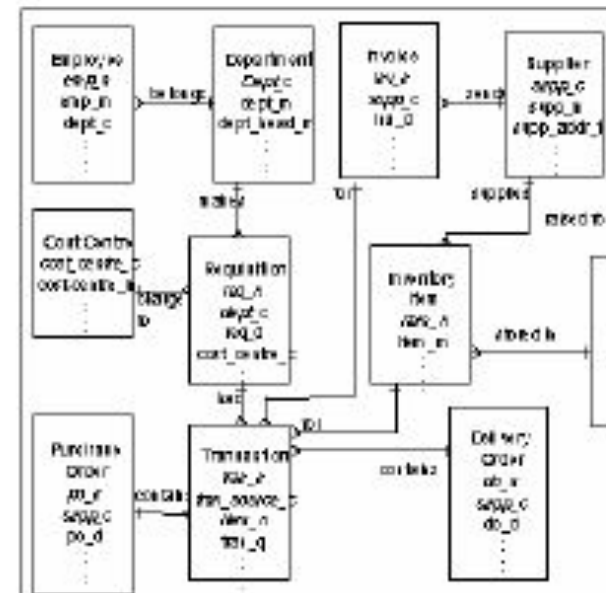
This is high level description of the structure of a database. E.g. E-R diagram

Concise description of the data requirements of the users and includes detailed descriptions of the data, relationships and constraints.

Database
Requirements



Conceptual Design



Database Design...

Conceptual Design¹



Phase 3 - Logical Design



Logical Design

This is the process of mapping the database structure developed in the previous phase to a particular database model.
E.g. map E-R model to relational

Specific to a database model,
but independent of a particular
DBMS (product)

ATTR1	ATTR2 A	TTTR3	ATTR4	ATTR	LL1.5
-------	---------	-------	-------	------	-------

Table

ATTR1	ATTR2 AT	TTR3	ATTR4
-------	----------	------	-------

Table

ATTR1	AT A TT R	4 ATTR5
-------	-----------	---------

Table

ATTR1	ATTR2 A	TTTR3	ATTR5
-------	---------	-------	-------

Table

Database Design...

Logical Design



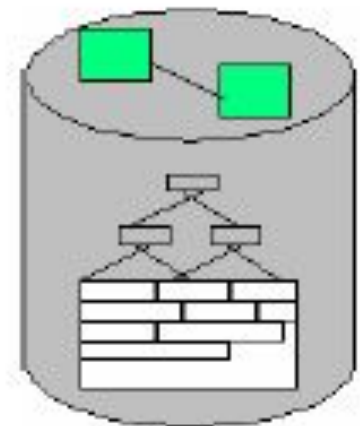
Phase 4 - Physical Design



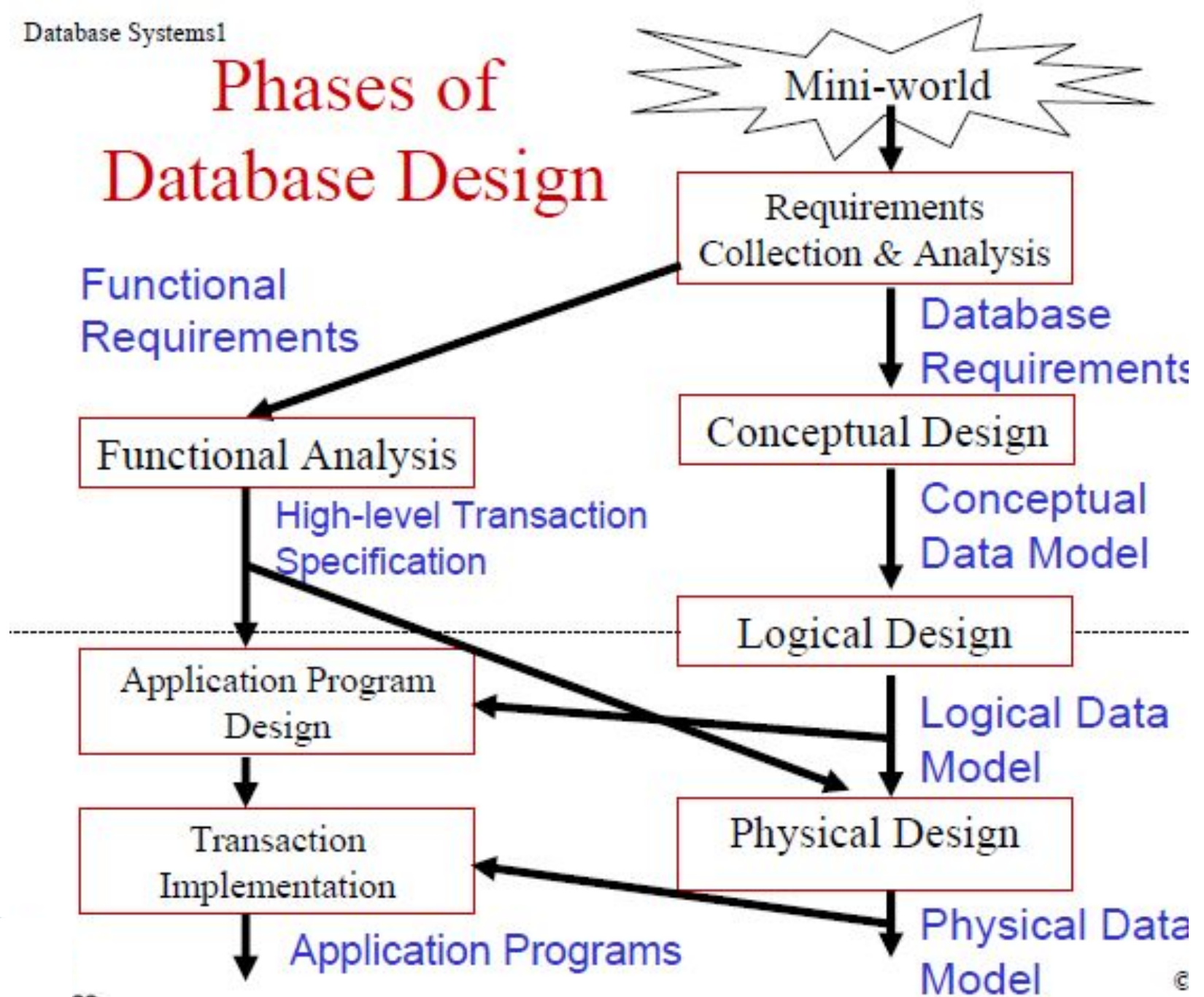
Physical Design

This is the process of defining structure that enables the database to be queried in an efficient manner.

E.g. index and hash file design, data partition



Phases of Database Design



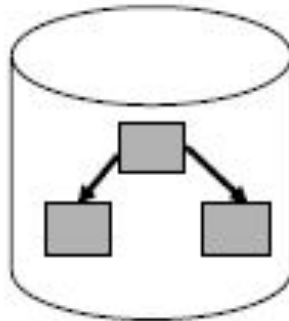
Types of Database Models

Traditional
Files



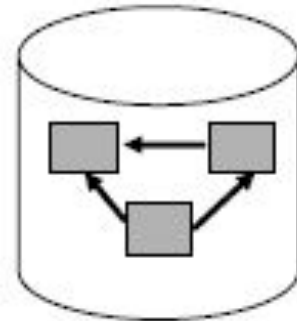
60s

Hierarchical
Database Model



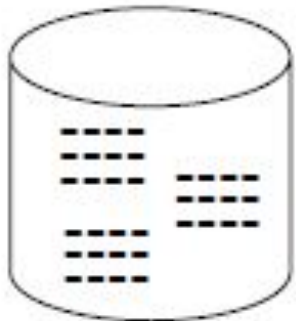
70s

Network
Database Model



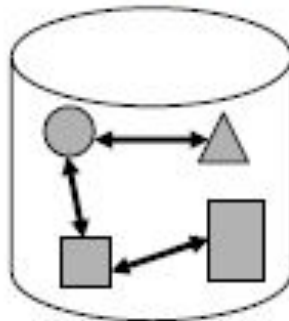
70s

Relational
Database Model



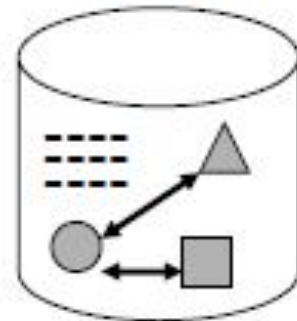
80s

Object-oriented
Database Model



90s

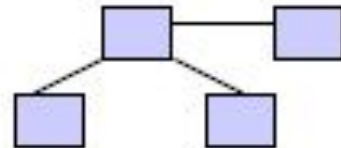
Object-relational
Database Model



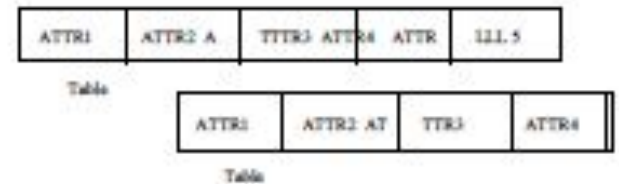
90s

Types of Data Models

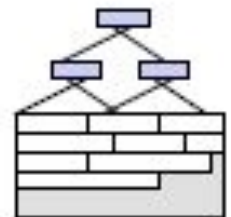
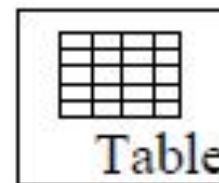
Conceptual Data Model



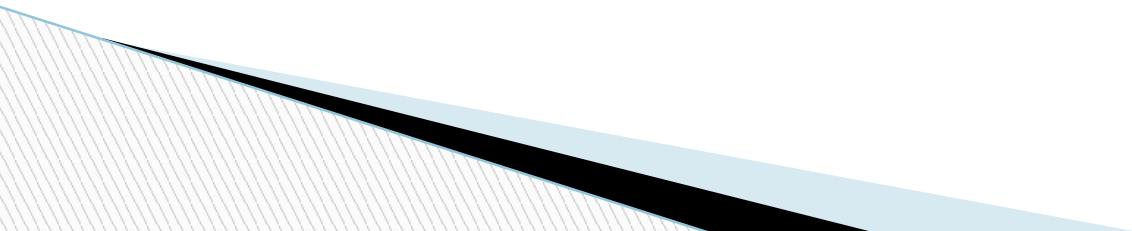
Logical Data Model



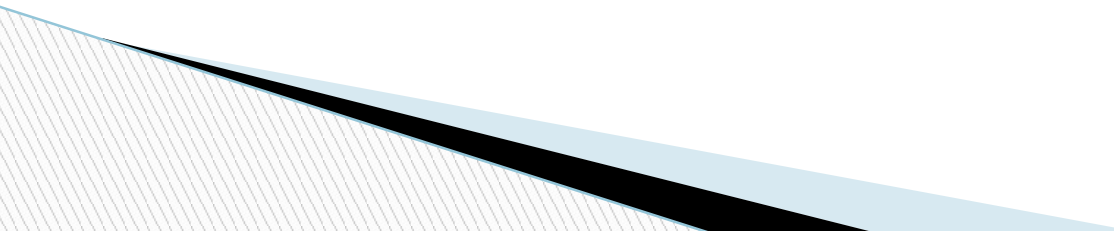
Physical Data Model



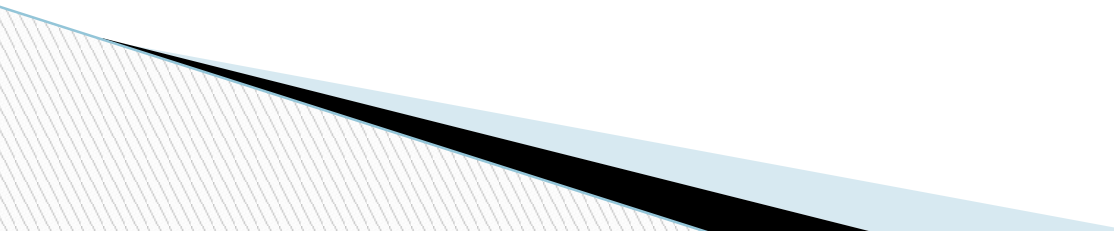
Conceptual Data Model

- A data model representing the objects and business rules that govern the operation of an organisation
 - *Done by a Business Analyst*
 - *Not constrained by access requirement and technology*
- 

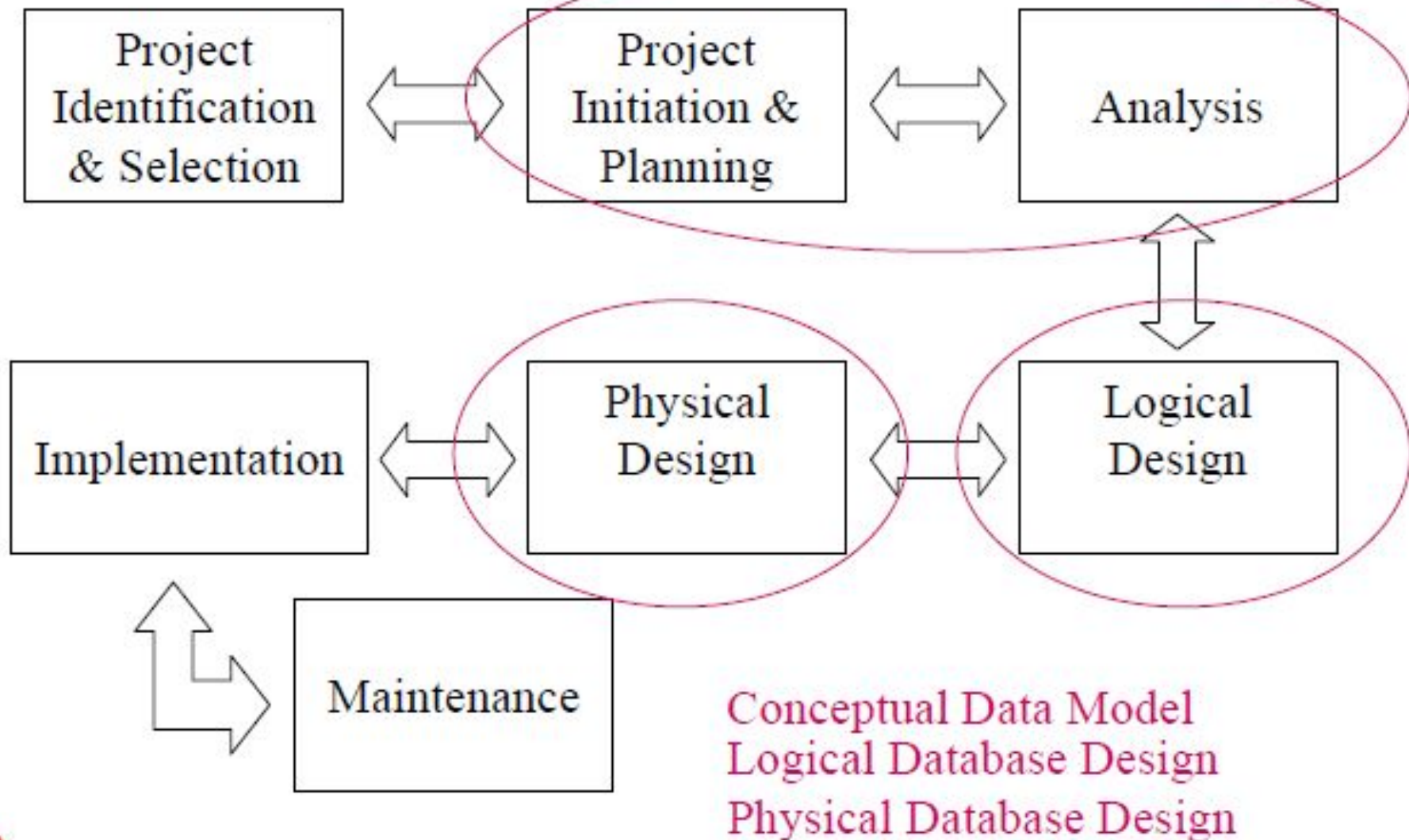
Logical Data Model

- ▣ A set of data structures assembled following rules that describe the processing requirements (access paths) of the data in terms of a logical database model
 - ▣ *Done by a Data Analyst*
 - ▣ *Not constrained by technology*
- 

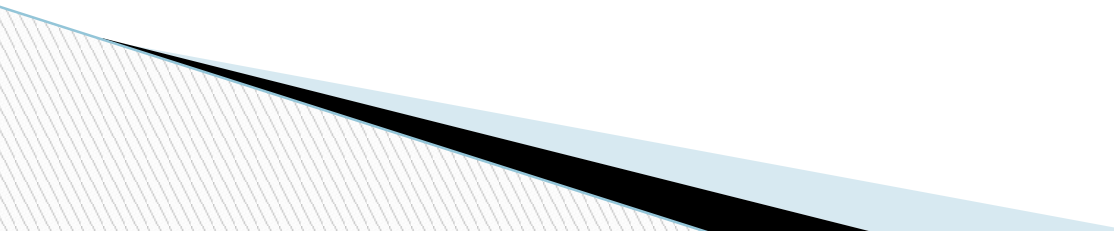
Physical Data Model

- ▣ A model prepared for the purpose of implementing a database that runs under the control of a particular DBMS (product)
 - ▣ *Done by a DBA*
 - ▣ *Constrained by Technology*
- 

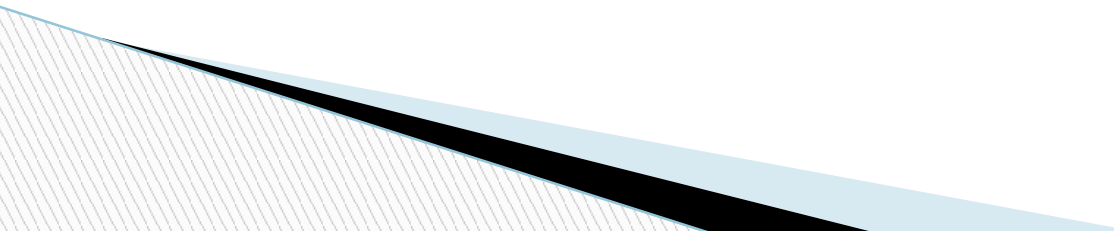
Systems Development Life Cycle



Database Development Activities

- Enterprise Modelling
 - Conceptual Data Modelling
 - Logical Database Design
 - Physical Database Design and Creation
 - Database Implementation
 - Database Maintenance
- 

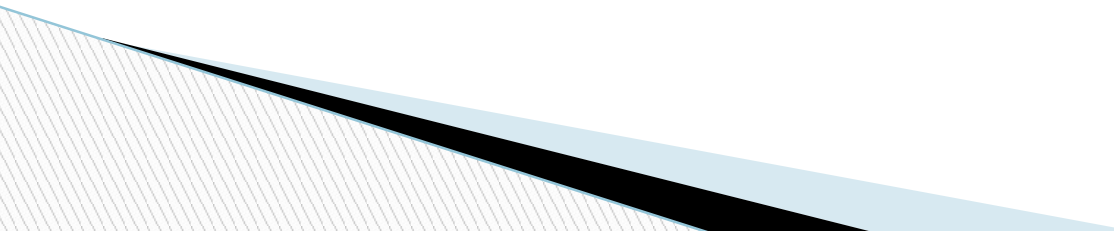
Enterprise Modelling

- Analyse current data processing
 - Analyse the general business functions and their database needs
 - Justify need for new data and databases in support of business
 - Project Identification & Selection
- 

Conceptual Data Modelling(Analysis)

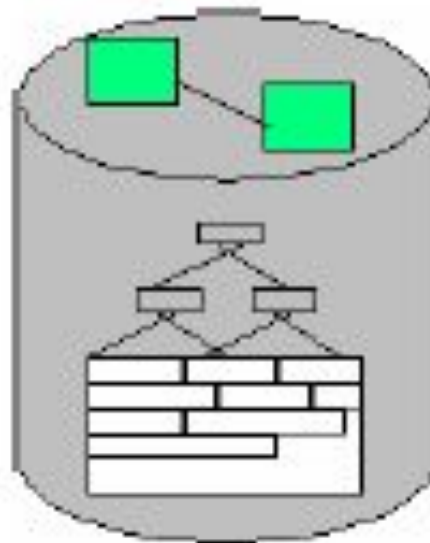
- Identify scope of database requirements for proposed information system
 - Analyse overall data requirements for business function(s) supported by database
 - Develop preliminary conceptual data model including entities and relationships
 - Compare preliminary conceptual data model with enterprise data model
 - Develop detailed conceptual data model, including all entities, relationships, attributes and business rules
 - Make conceptual data model consistent with other models of information system
 - Populate repository with all conceptual database specifications
- 

Logical Database Design(Logical Design)

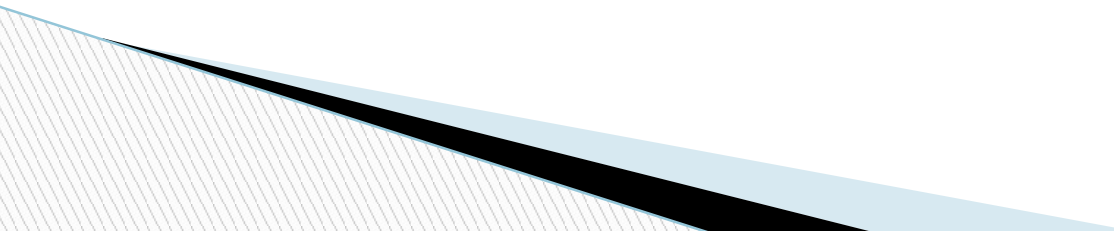
- Analyse in detail the transactions, forms, displays and inquires (data views) required by the business functions supported by the database
 - Integrate database views into conceptual data model
 - Identify data integrity and security requirements, and populate repository
- 

Physical Database Design

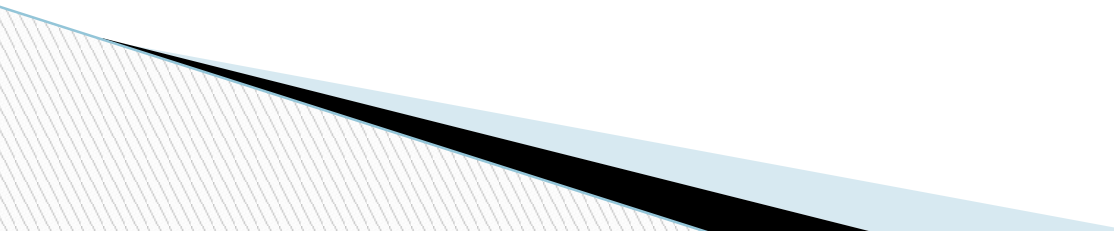
- Define database to DBMS (often generated from repository)
- Decide on physical organisation of data
- Design database processing programs



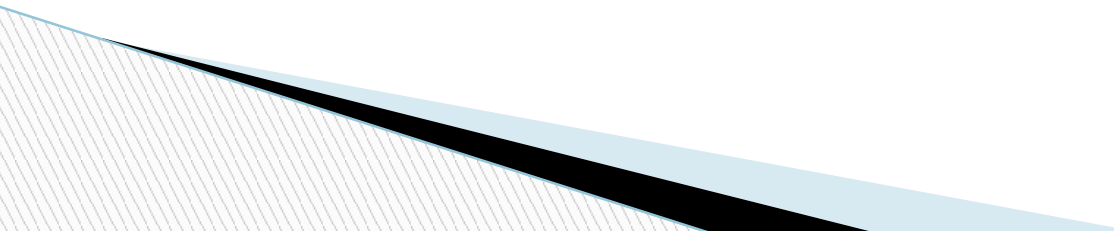
Database Implementation

- Code and test database processing programs
 - Complete database documentation and training materials
 - Install database and convert data from prior systems
- 

Database Maintenance

- Analyse database and database applications to ensure that evolving information requirements are met
 - Tune database for improved performance
 - Fix errors in database and database applications and recover database when it is contaminated
- 

Conceptual Design

- All the requirements collected at *Phase 1* are analysed to create a *Conceptual Schema*. This process is called the *Conceptual Design*. We identify the *entities, their attributes, relationships and constraints (business rules)*.
 - The conceptual schema is used as a reference to ensure that all user's data requirements are met and the requirements do not include any conflicts.
- 

Conceptual Design

Entities

Department



Employee



Project



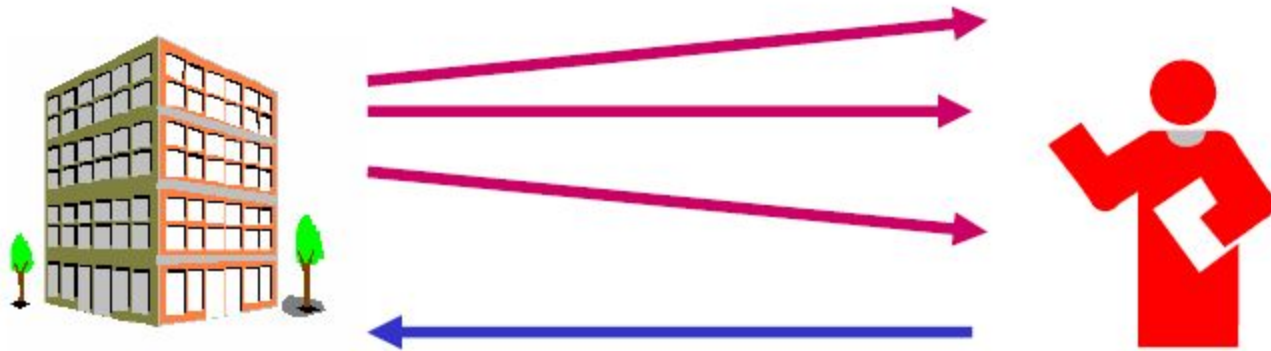
Dependent



Conceptual Design

Relationships

A Department has **Many Employees**



An Employee works for **A Department**

Relationships

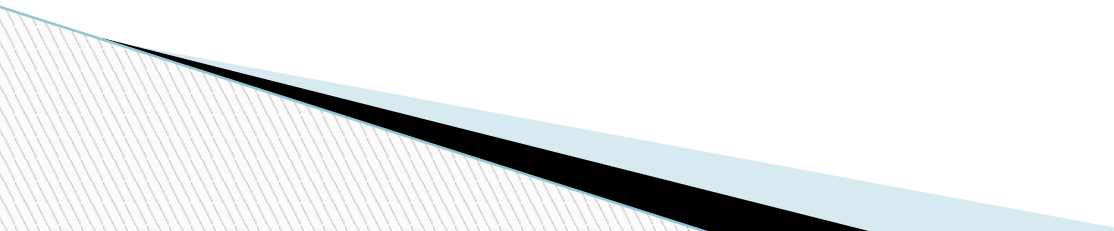
□ Relationship Type

- A meaningful association between association between (or among) entity types

□ Relationship Instances

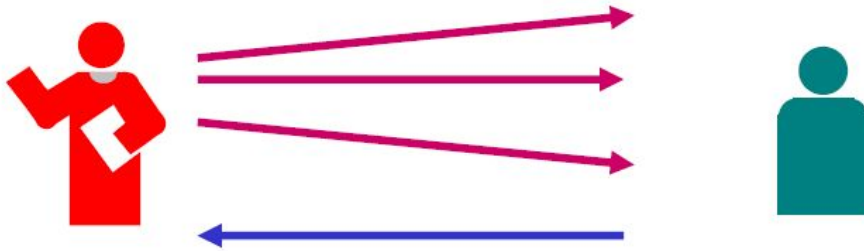
- An association between (or among) entity instances, where each relationship instance includes exactly one entity from each participating entity type

e.g. De Silva works for Personnel Department



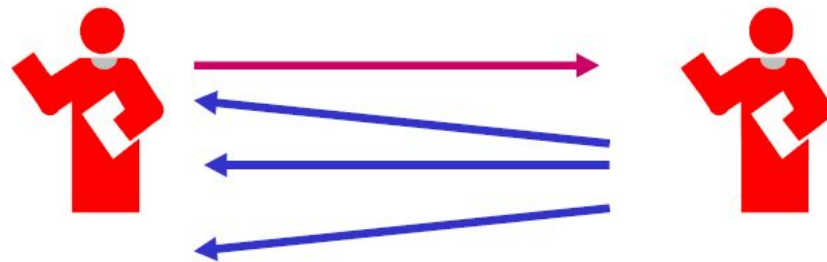
Relationships

An Employee has Many Dependants



A Dependant belongs to An Employee

An Employee supervised by An Employee

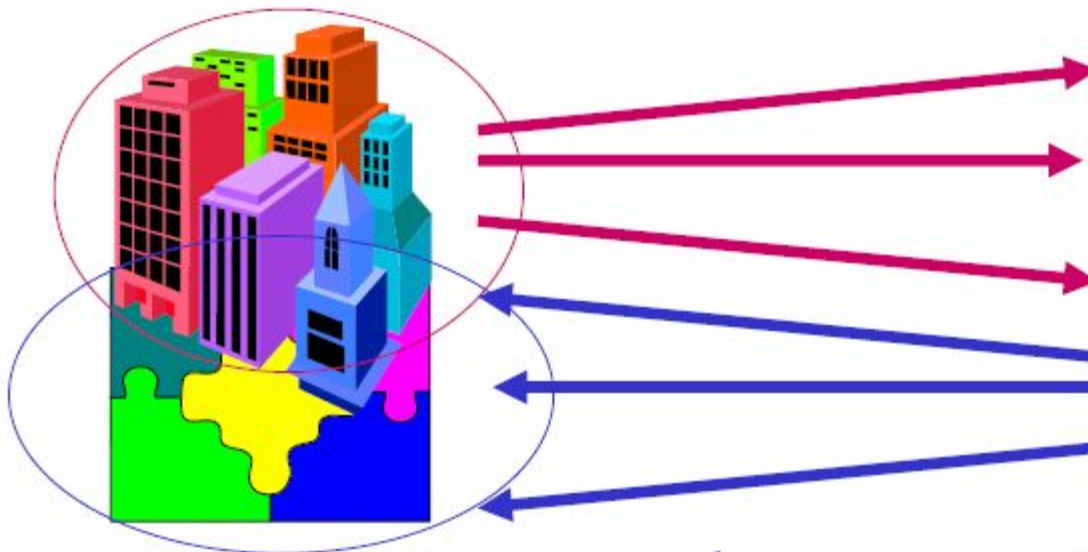


An Employee supervise Many Employees

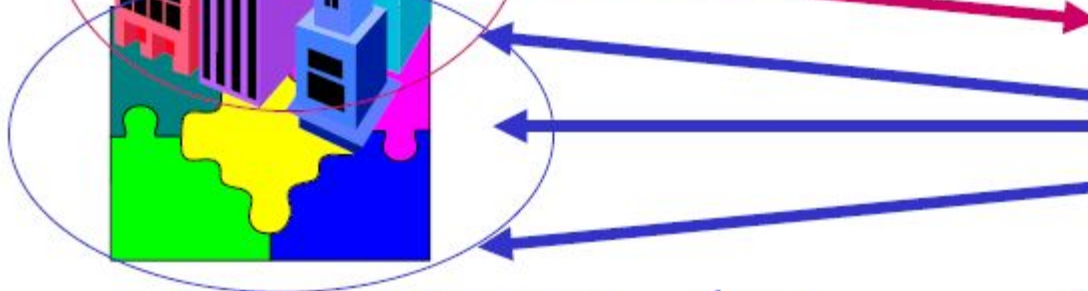
Conceptual Design

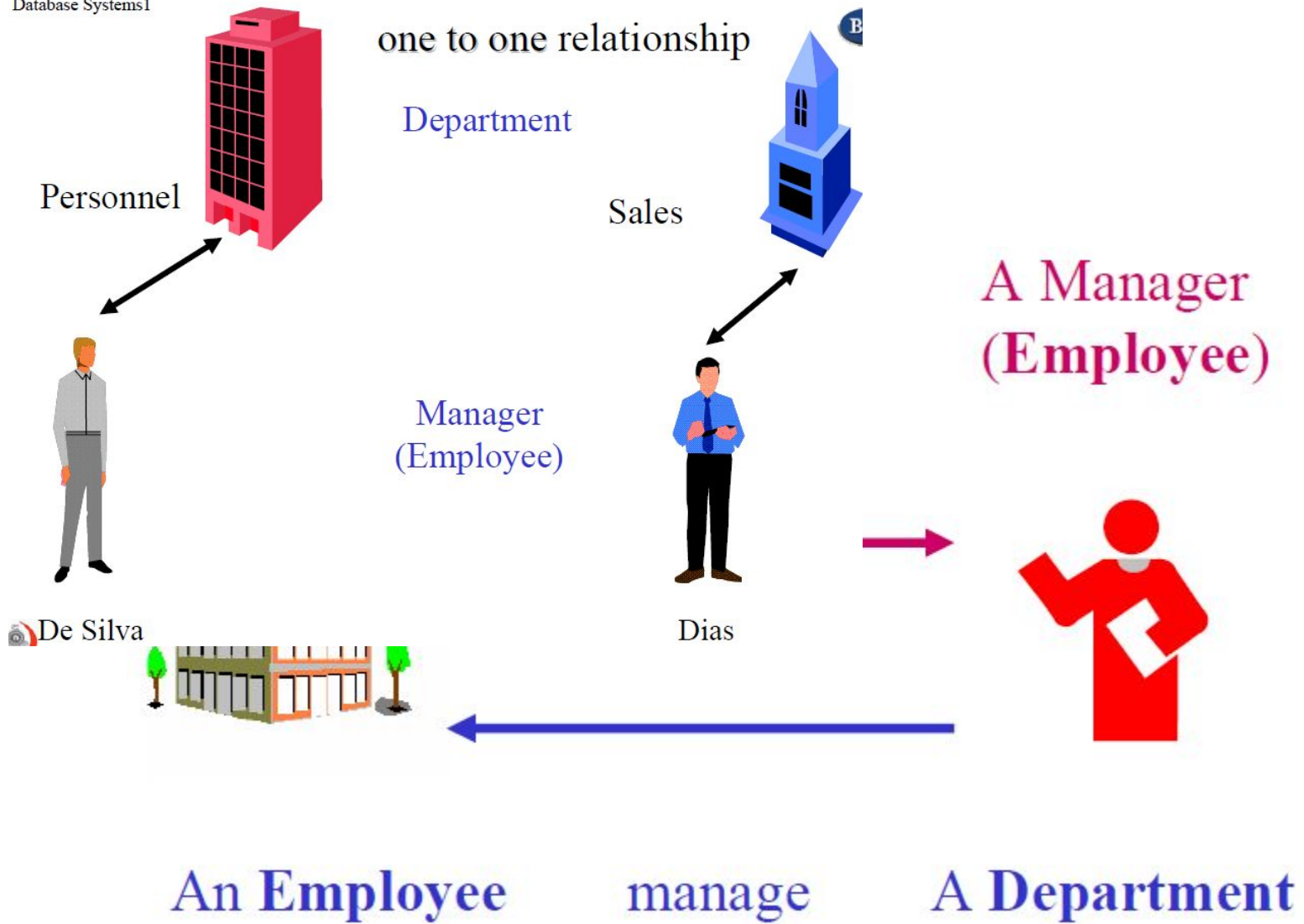
Entities / Relationships??

A Company has **Many Departments**

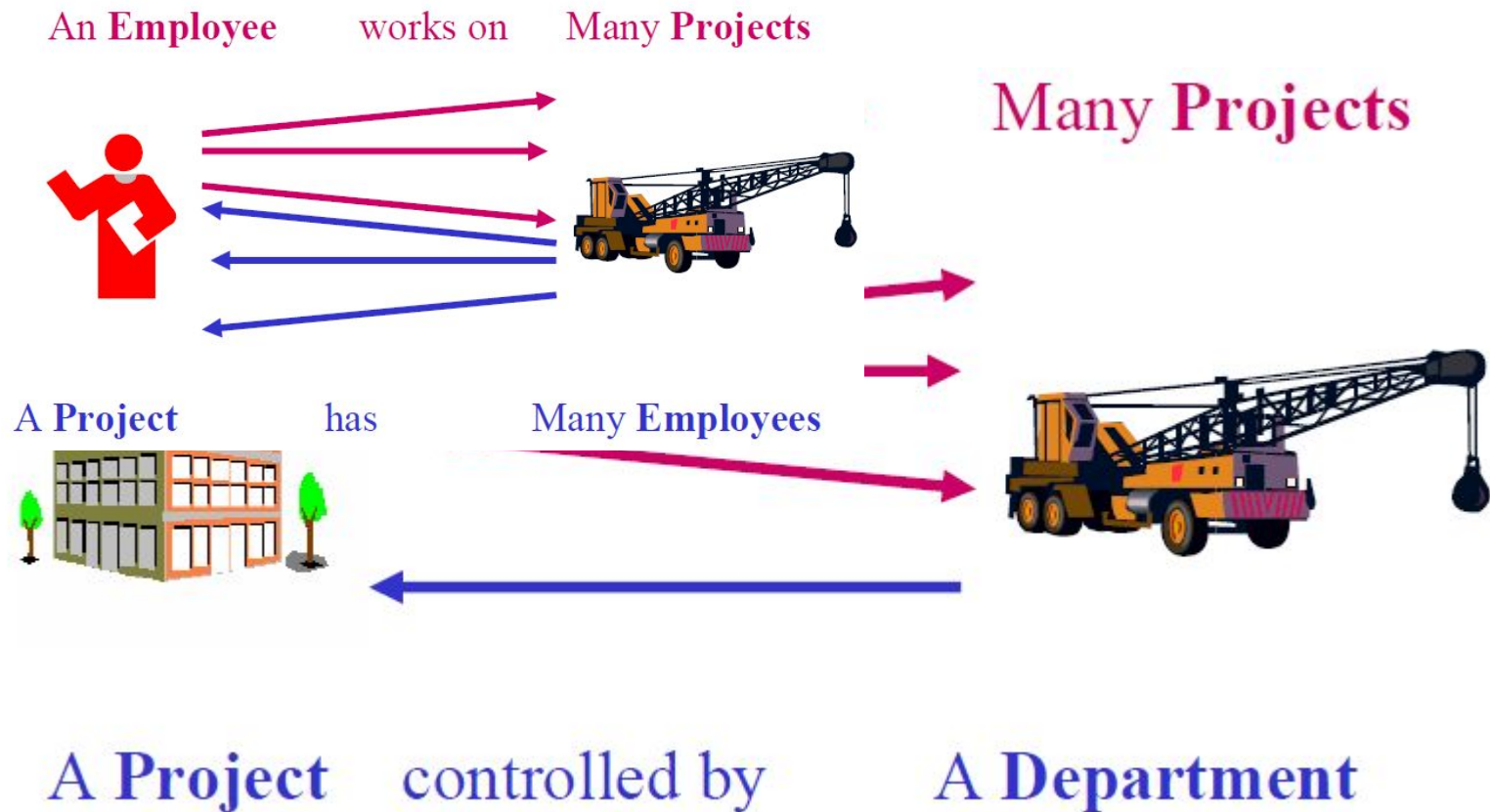


A Department has **Many Locations**

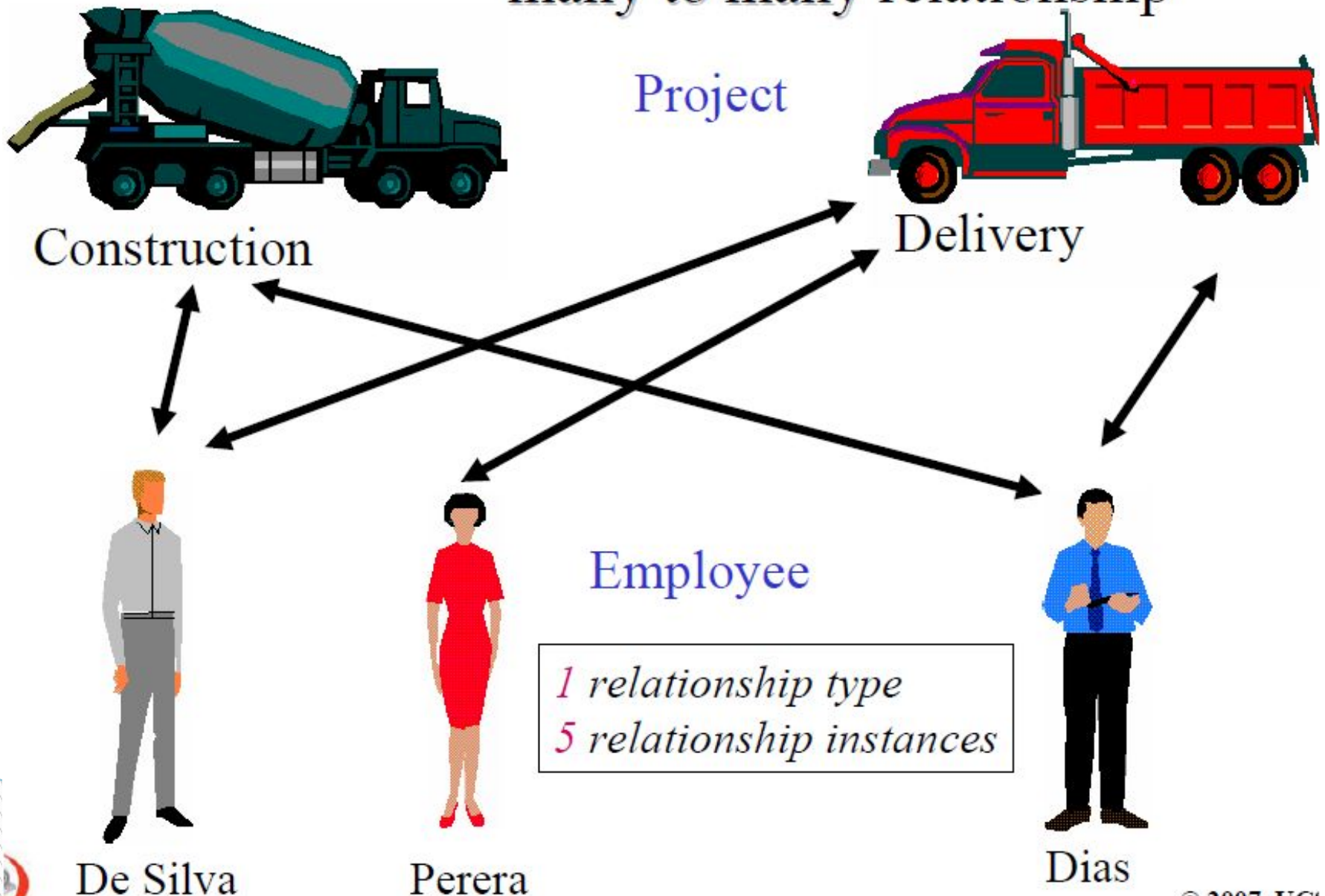




One to Many Relationship



many to many relationship



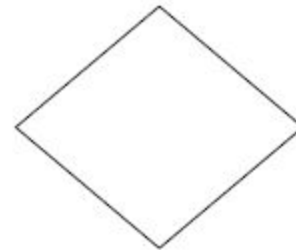
Conceptual Design

▣ Notations

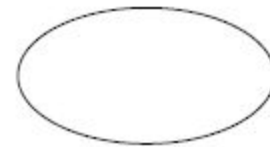
Entity



Relationship



Attribute



Relationship Types

One to One



One to Many



Many to Many



Cardinality Constraints

- Specifies the number of instances of one entity that can (or must) be associated with each instance of another entity
- Minimum Cardinality
 - The minimum number of instances of one entity that may be associated with each instance of another entity

e.g. the minimum dependants for an employee is zero

Cardinality Constraints

- Optional Participation
 - when the number of participants in the relationship is zero
- Mandatory Participation
 - when the number of participants in the Relationship is one
- Maximum Cardinality
 - The maximum number of instances of one entity that may be associated with a single occurrence of another entity

e.g. an Employee can have insurance policies for at most two dependants (0:2)

Existence Conditions

One to One



One to Many

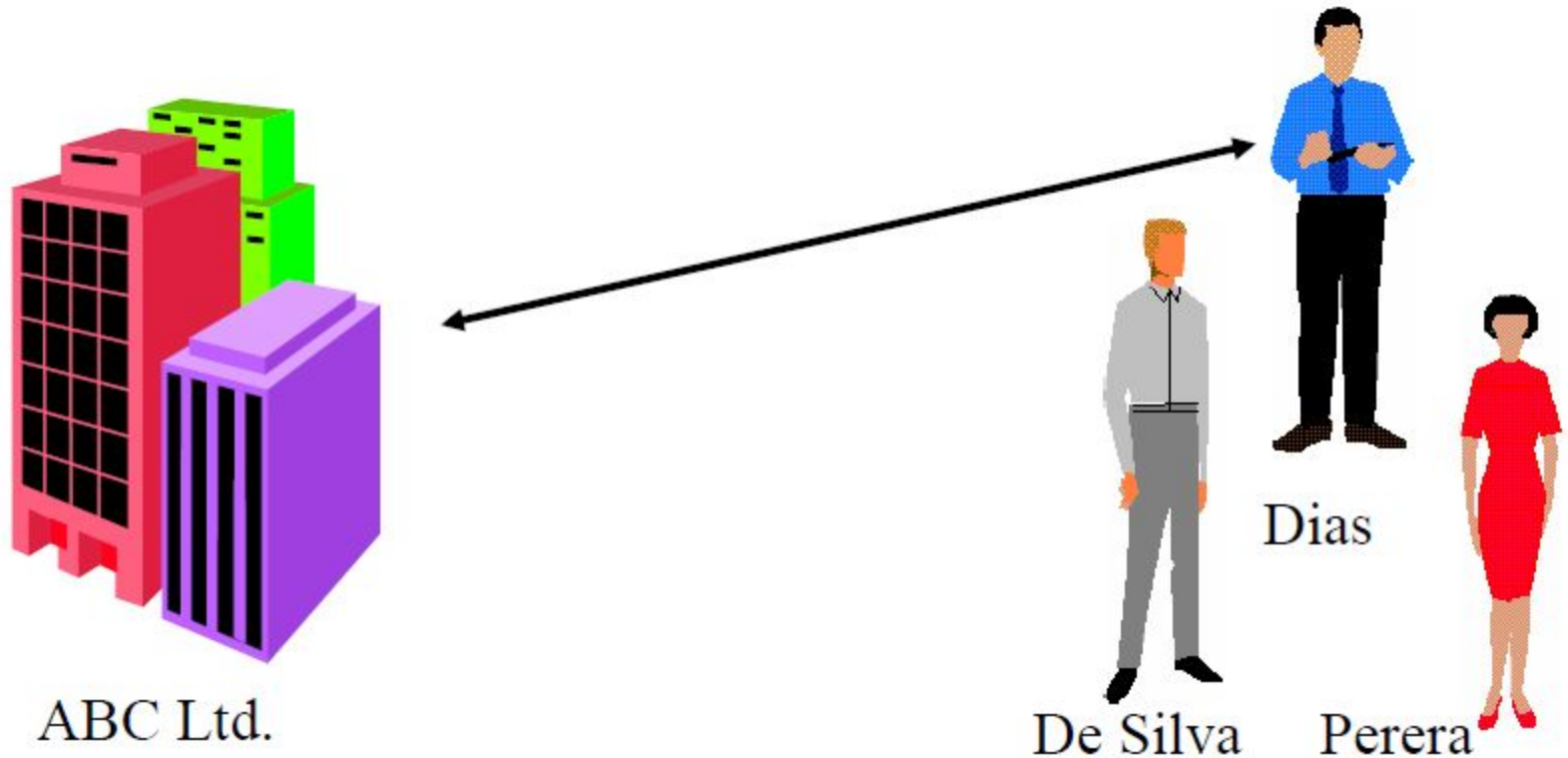


Many to Many



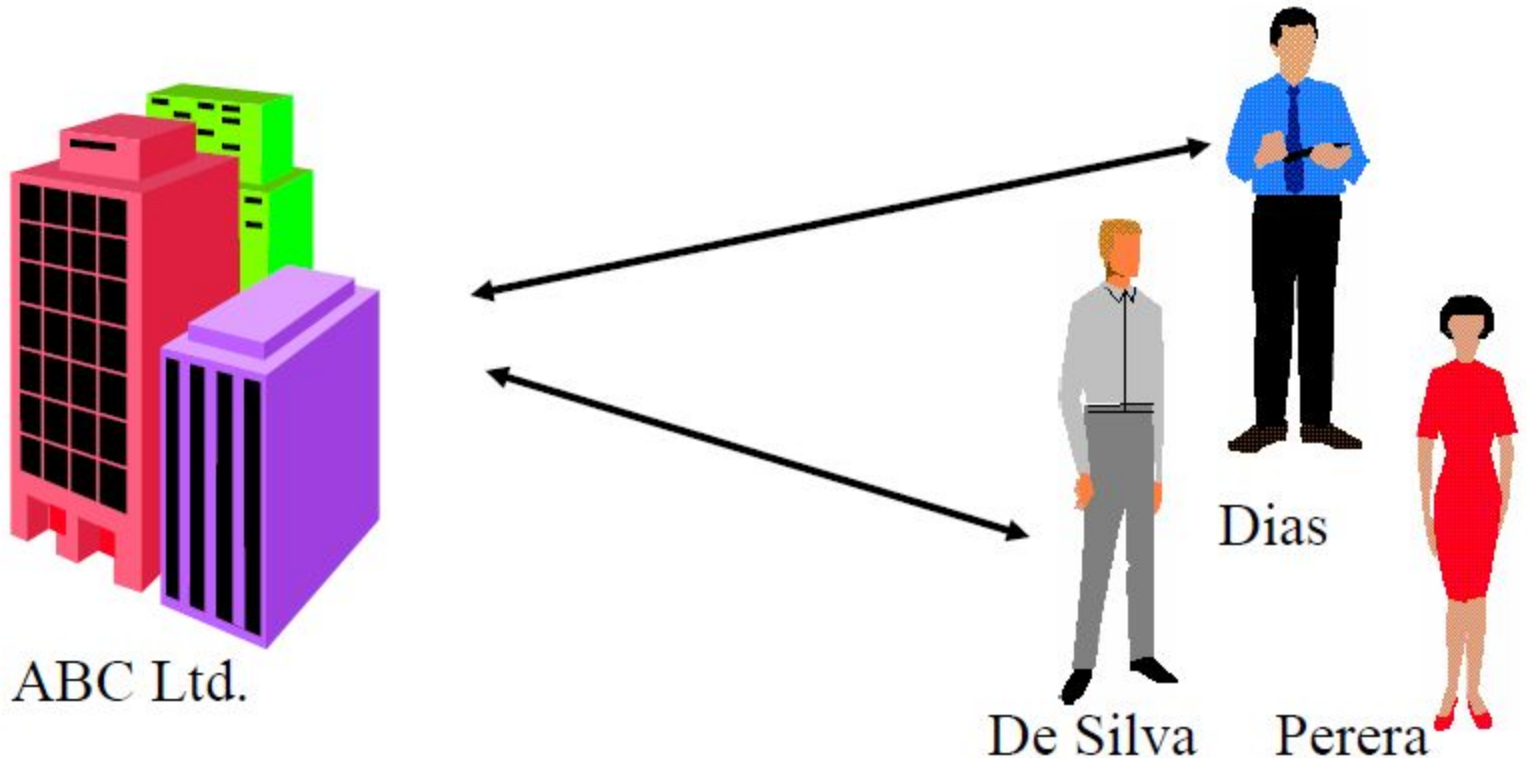
Existence Conditions

One to One



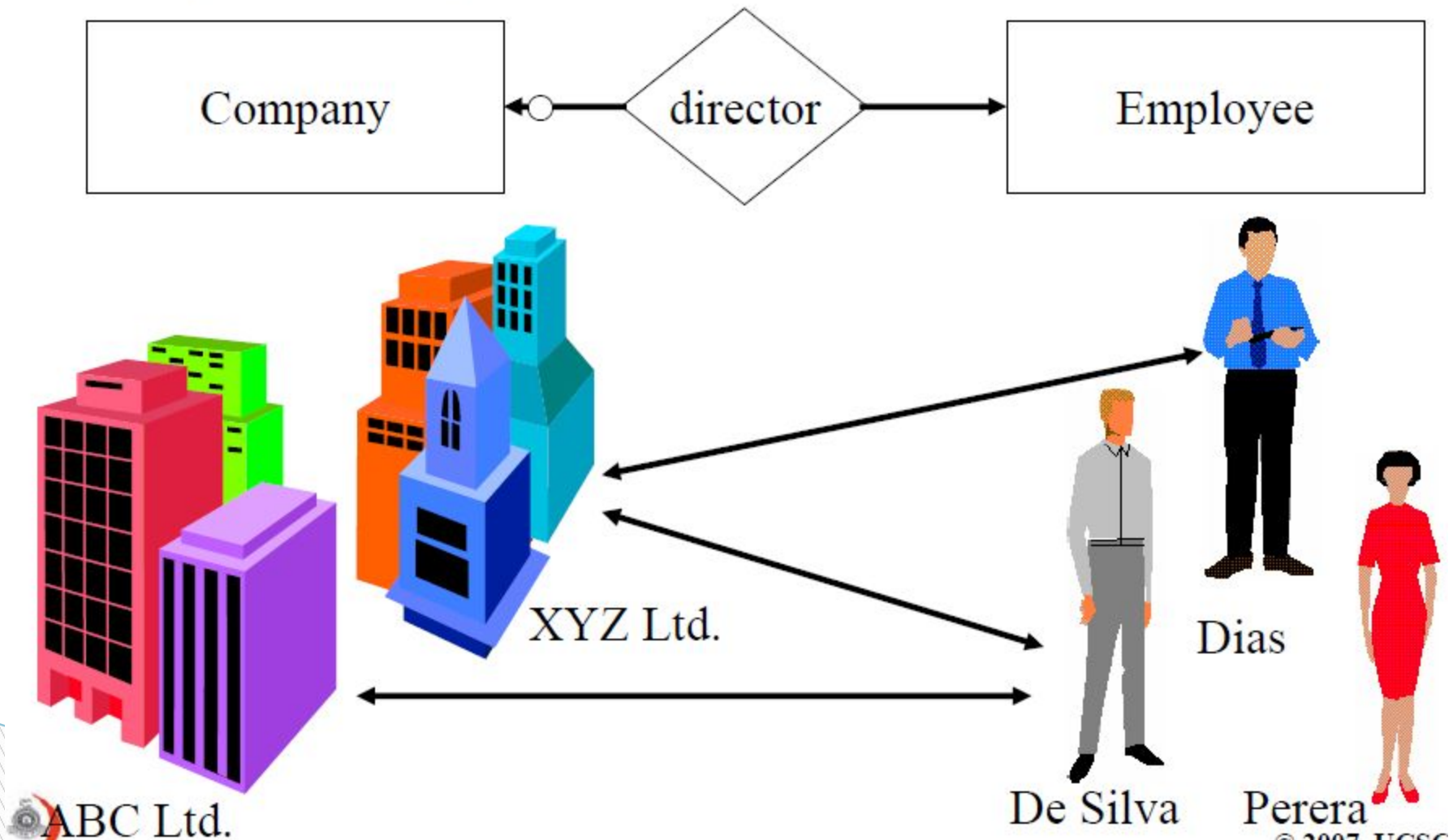
Existence Conditions

One to Many



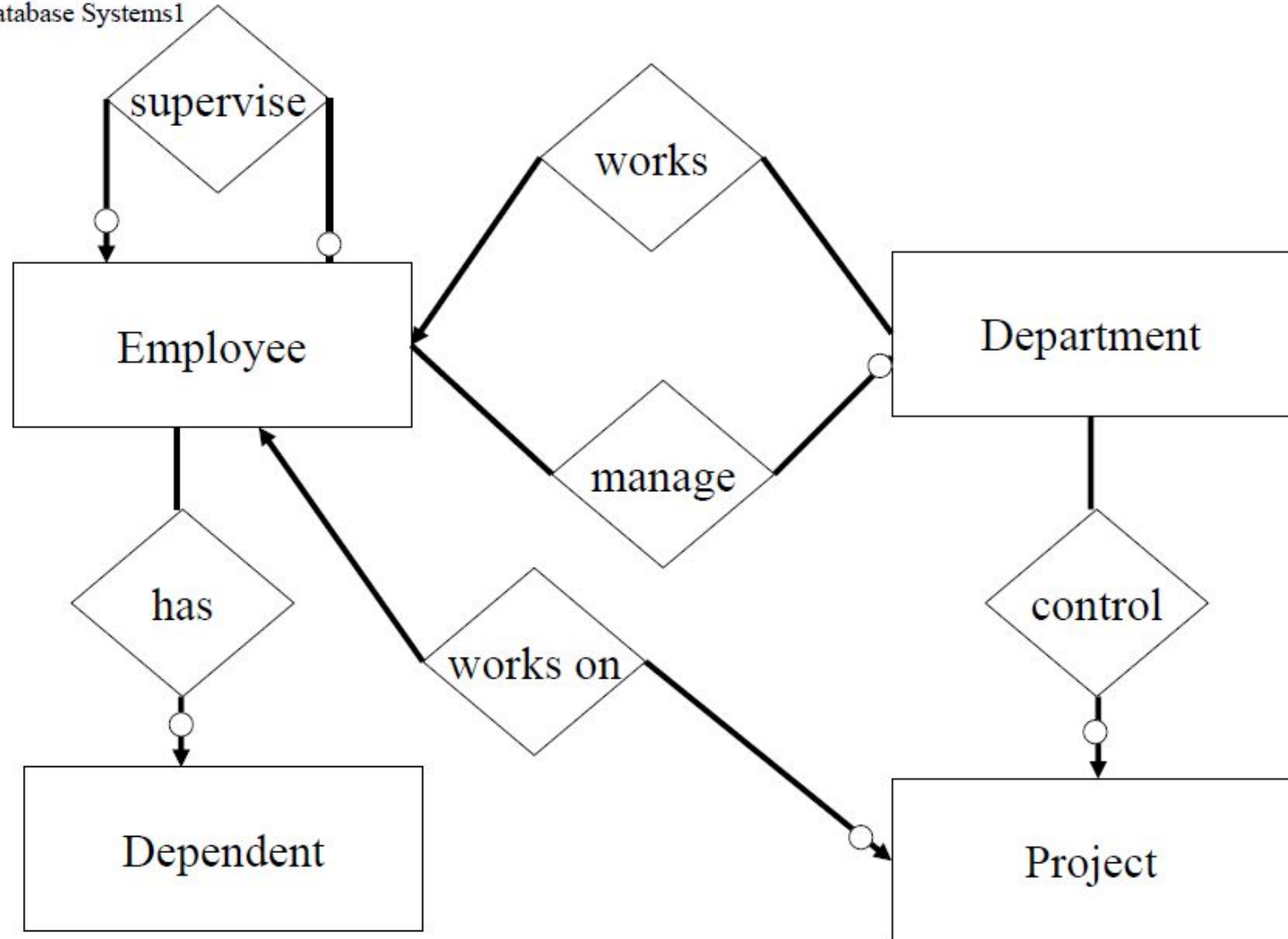
Existence Conditions

Many to Many



Existence Conditions

Database Systems1

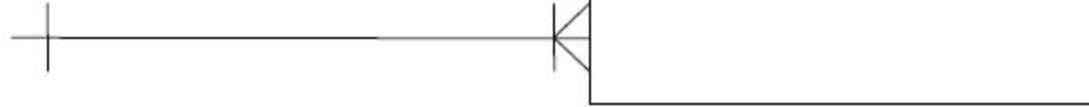


E-R Modelling

1:1 Ma



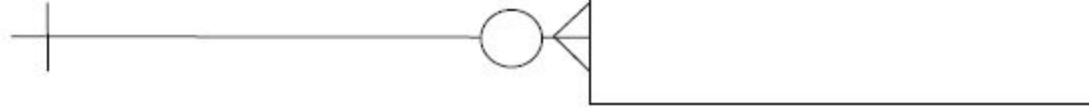
1:N Ma



1:1 Op



1:N Op



Martin (Information Engineering)

E-R Modelling

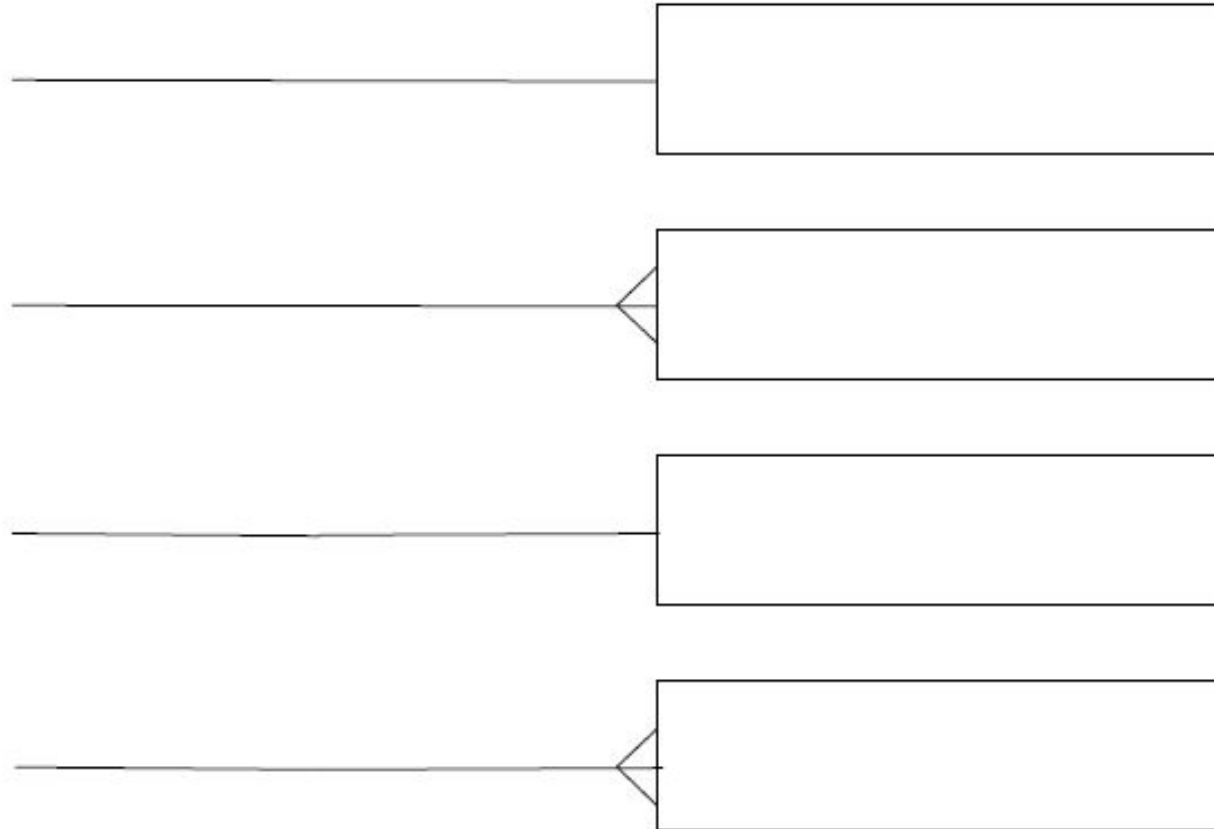
1:1 Ma

1:N Ma

1:1 Op

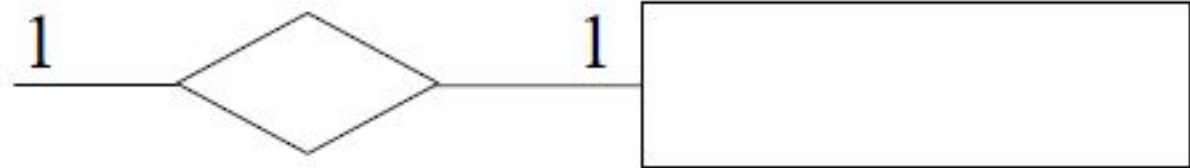
1:N Op

Crow's Foot



E-R Modelling (Alternate Notations)

1:1 Ma



1:N Ma



Optional not shown

E-R Modelling (Alternate Notations)

1:1 Ma

1,1

1,1

A diagram showing a 1:1 Mandatory relationship. A horizontal line connects the cardinality '1,1' on the left to the cardinality '1,1' on the right. A vertical line extends from the midpoint of this horizontal line to a rectangular box on the right.

1:N Ma

1,N

1,1

A diagram showing a 1:N Mandatory relationship. A horizontal line connects the cardinality '1,N' on the left to the cardinality '1,1' on the right. A vertical line extends from the midpoint of this horizontal line to a rectangular box on the right.

1:1 Op

0,1

1,1

A diagram showing a 1:1 Optional relationship. A horizontal line connects the cardinality '0,1' on the left to the cardinality '1,1' on the right. A vertical line extends from the midpoint of this horizontal line to a rectangular box on the right.

1:N Op

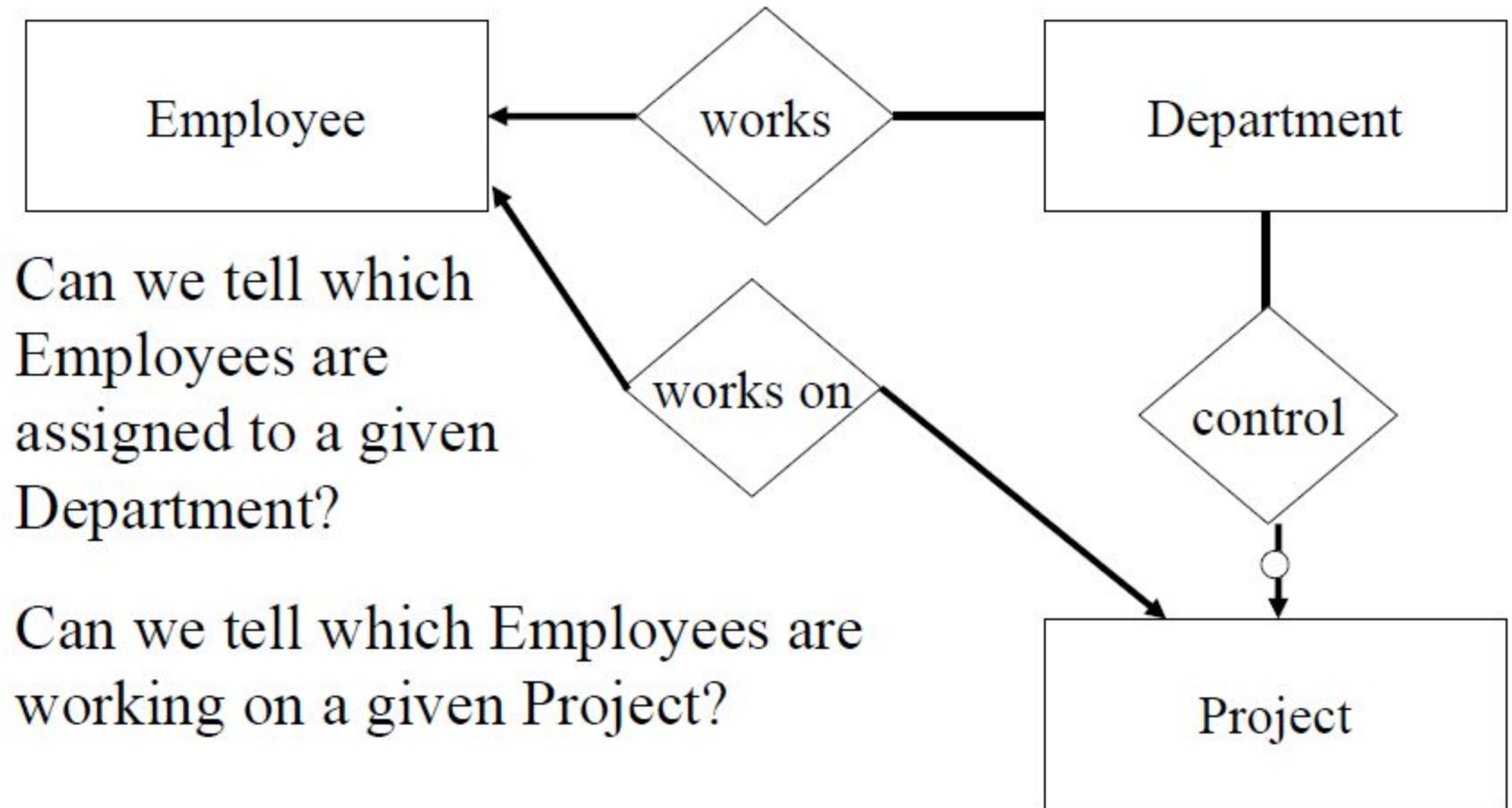
0,N

1,1

A diagram showing a 1:N Optional relationship. A horizontal line connects the cardinality '0,N' on the left to the cardinality '1,1' on the right. A vertical line extends from the midpoint of this horizontal line to a rectangular box on the right.

Datarun

Using Relationship to Define Access Paths



Can we tell which Employees are assigned to a given Department?

Can we tell which Employees are working on a given Project?

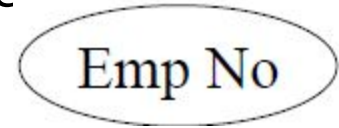
Can we tell which Employee work on Projects that do not belong to their Department?

Detail Conceptual Data Model

Attributes

□ Attribute

- A property or characteristic of an entity type ^{that is of} interest to the organisation



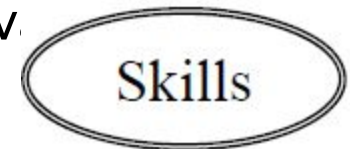
□ Simple Attribute

- An attribute that cannot be broken down into smaller components

□ Multi-valued Attribute

- An attribute that may take on more than one v given entity instance

e.g. Employee Skills, Qualifications

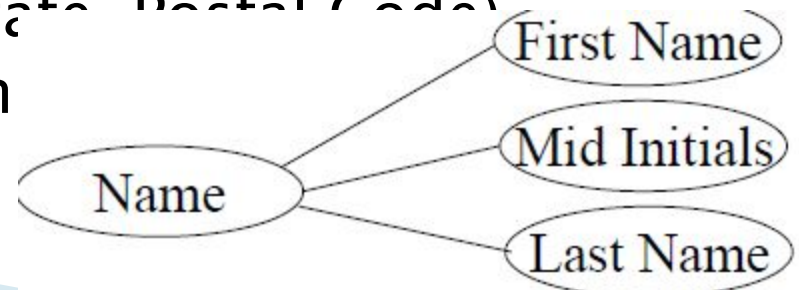


□ Composite Attribute

- An attribute that can be broken down into component parts

e.g. Address (Street, City, State, Postal Code)

Name (First Name, Middle In



Attributes Cont'd

□ Stored Attribute

- An attribute whose value is stored in the database

□ Derived Attribute

- An attribute whose values can be calculated from related attribute values

e.g. Years Employed (using Employed Date)

Age (using Date of Birth)

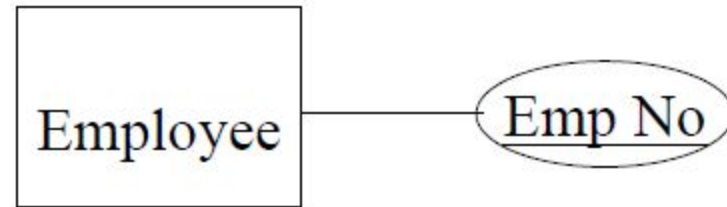


Age

Identifier

□ Identifier (Unique attribute)

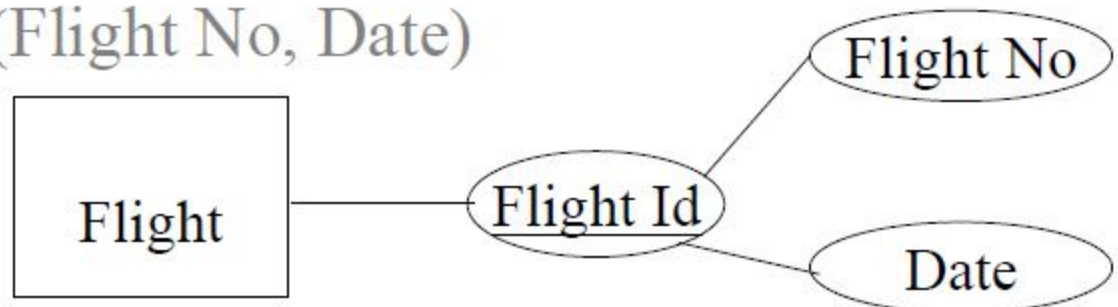
- An attribute (or combination of attributes) that uniquely identifies individual instances of an entity type. e.g. Emp No



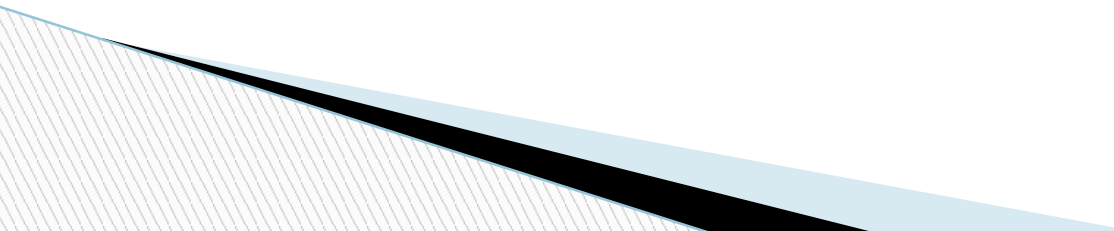
□ Composite Identifier

- An identifier that consists of a composite attribute

e.g. Flight Id (Flight No, Date)

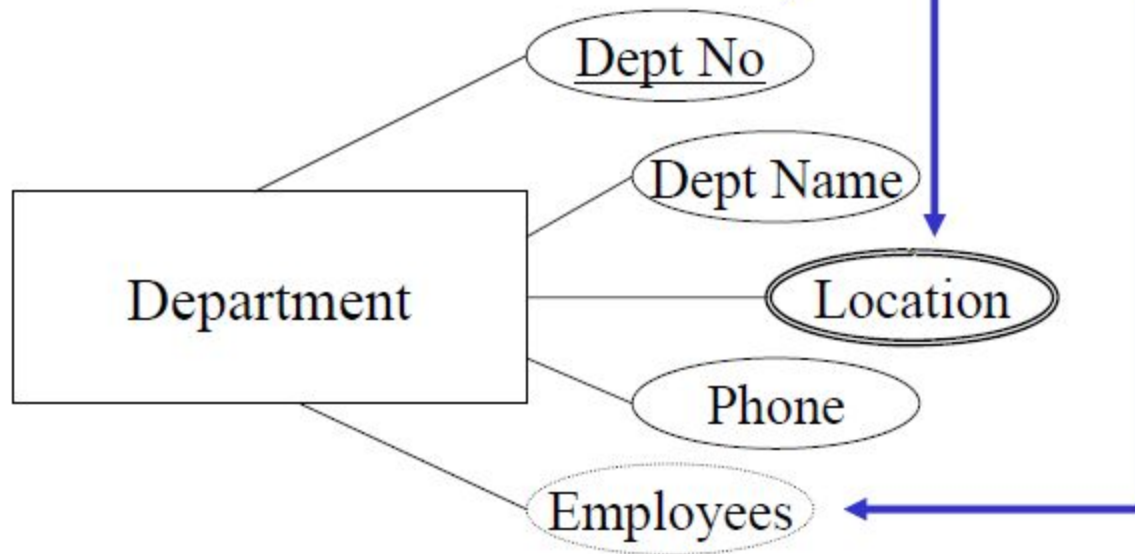


Identifier

- ❑ Choose an identifier that will not change its value over the life of each instance of the entity type
 - ❑ Choose an identifier such that each instance of the entity type, the attribute is guaranteed to have valid values and not be null (or unknown)
 - ❑ Avoid the use of so-called intelligent identifiers, whose structure indicates classifications, etc.
 - ❑ Consider substituting single-attribute identifiers for large composite identifiers
- 

Detailed Conceptual Design

Dept No	unique identifier of a dept.	Identifier
Dept Name	name of a department	Unique
Location	location of a department	Multi-valued
Phone	phone no. of a department	
Employees	no. of employees in a dept.	Derived



Detailed Conceptual Design

Project

Name

Department **Control**

Number

Location

Leader → *Employee*

Proj No

unique identifier of a project

Identifier

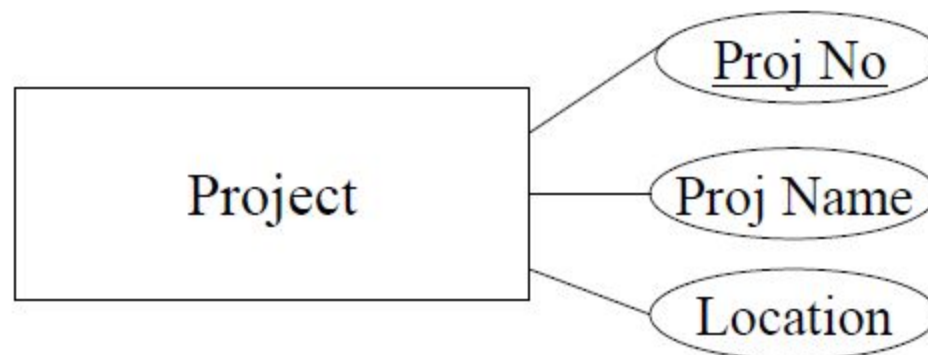
Proj Name

name of a project

Unique

Location

location of a project



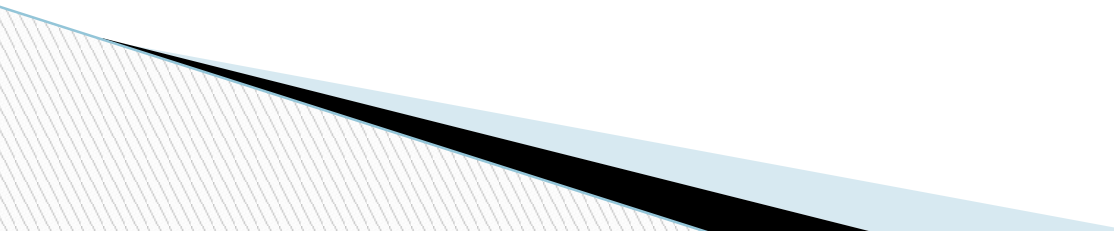
Detailed Conceptual Design

Employee

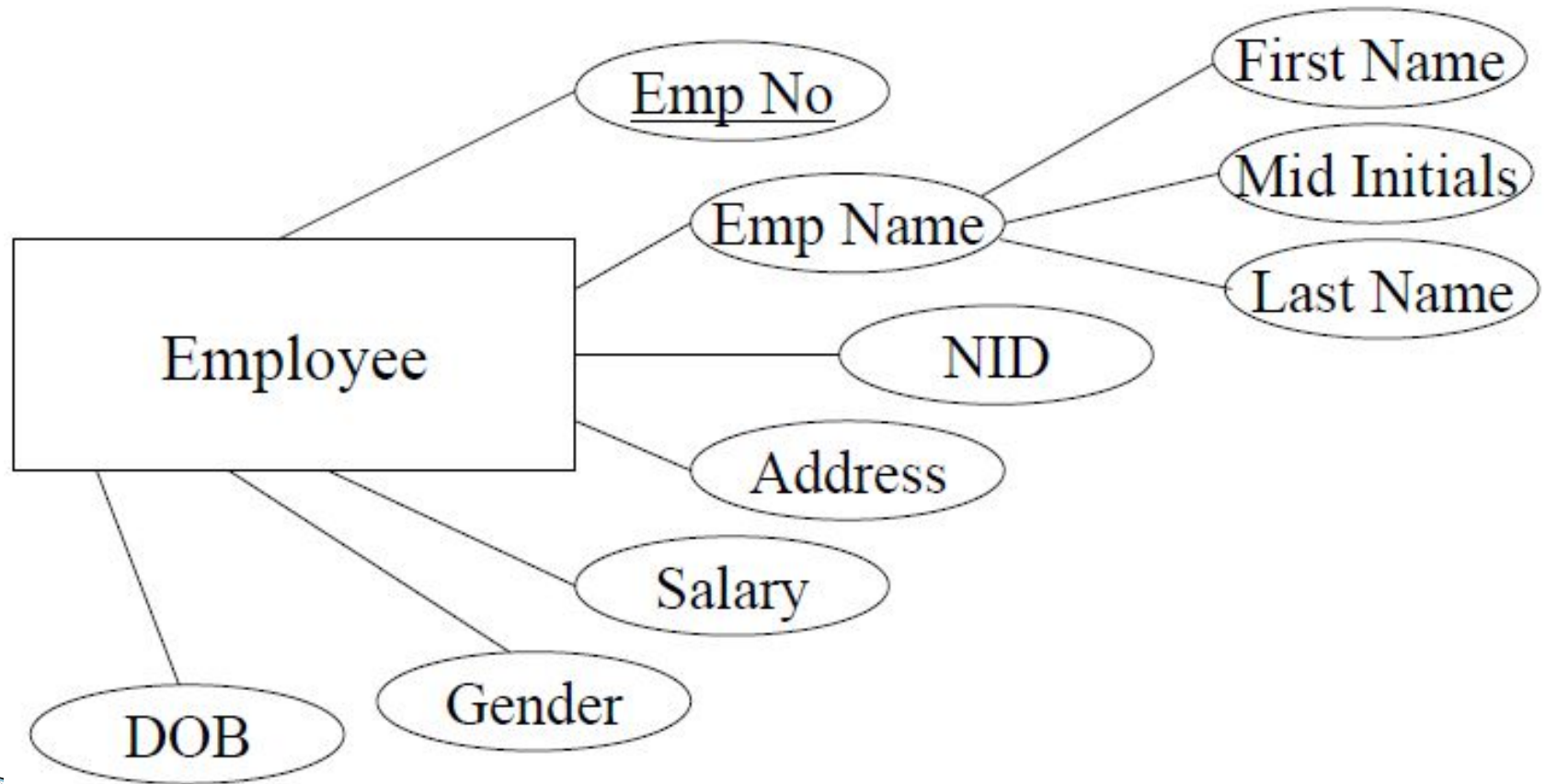


Emp No	unique identifier of an emp.	Identifier
Emp Name	name of an employee	Composite
First Name	first name of an employee	
Mid Initials	middle initials of an employee	
Last Name	last name of an employee	
NID	national id of an employee	Unique
Address	address of an employee	
Salary	salary of an employee	
Gender	sex of an employee	
DOB	birth date of an employee	

Detailed Conceptual Design

- Employee
 - Emp No
 - Name
 - National ID
 - Address
 - Salary
 - Sex
 - Birth Date
 - Works for Department
 - Supervise Employee
- 

Detailed Conceptual Design

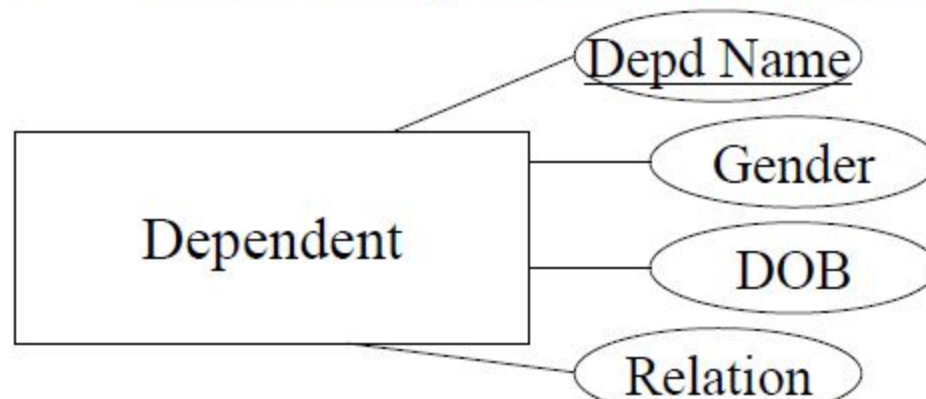


Detailed Conceptual Design

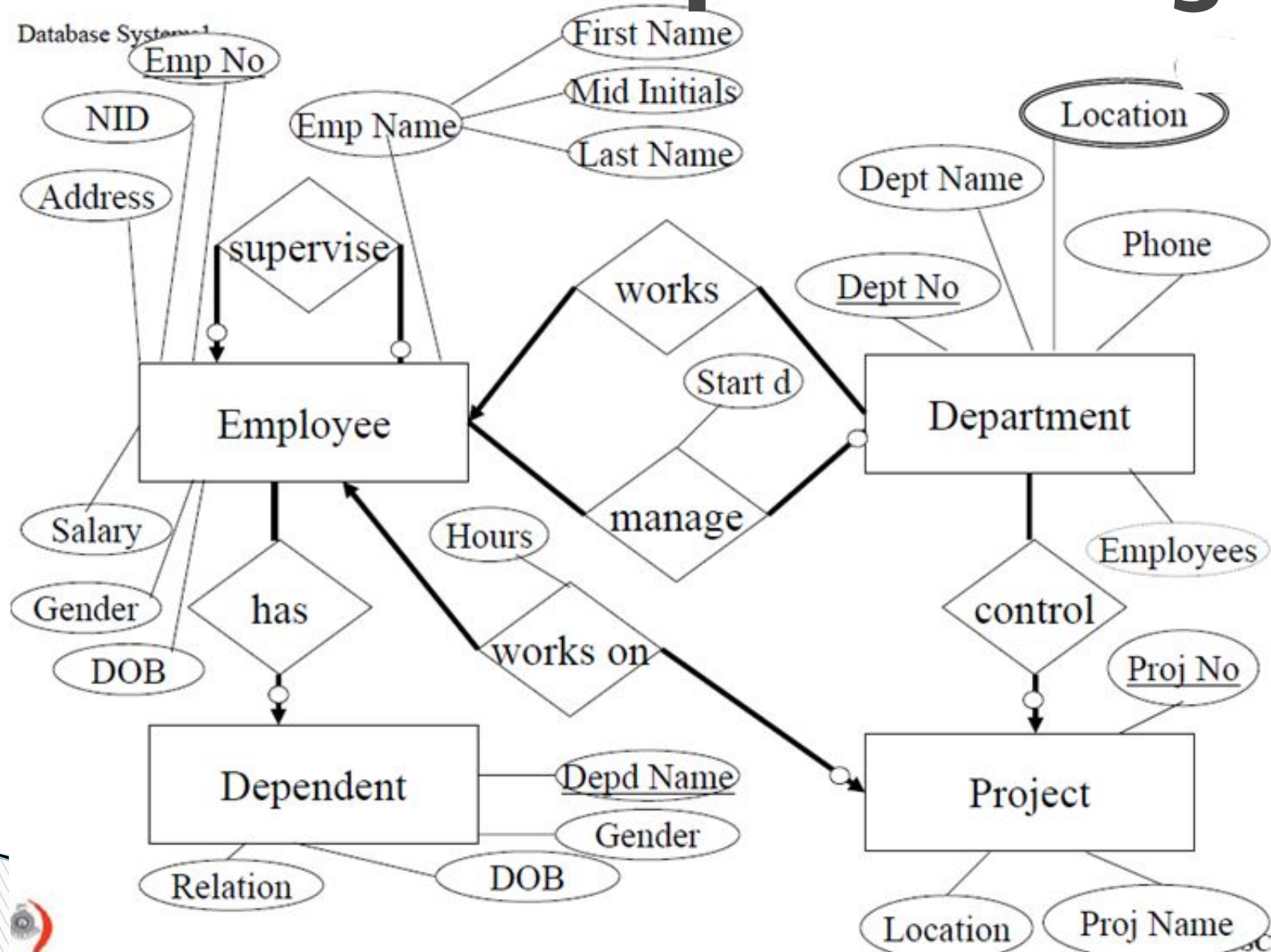
Dependent

Name	Sex	Birth Date	Relationship
------	-----	------------	--------------

Depd Name	name of a dependent	Part of Key
Gender	sex of a dependent	
DOB	birth date of a dependent	
Relation	relationship of a dependent to an employee	



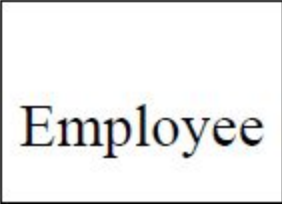
Detailed Conceptual Design



Entity Types

□ Strong (Regular) Entity

- An entity that exists independently of other entity types

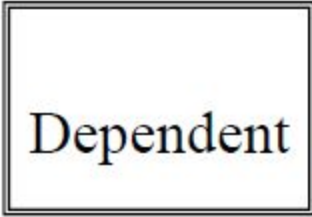


Employee

A rectangular box with a single border, representing a strong entity.

□ Weak Entity

- An entity types whose existence depends on some other entity



Dependent

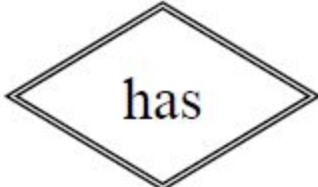
A rectangular box with a double border, representing a weak entity.

□ Identifying Owner

- The entity type on which the weak entity type depends
e.g. Employee is the Owner of Dependent

□ Identifying Relationship

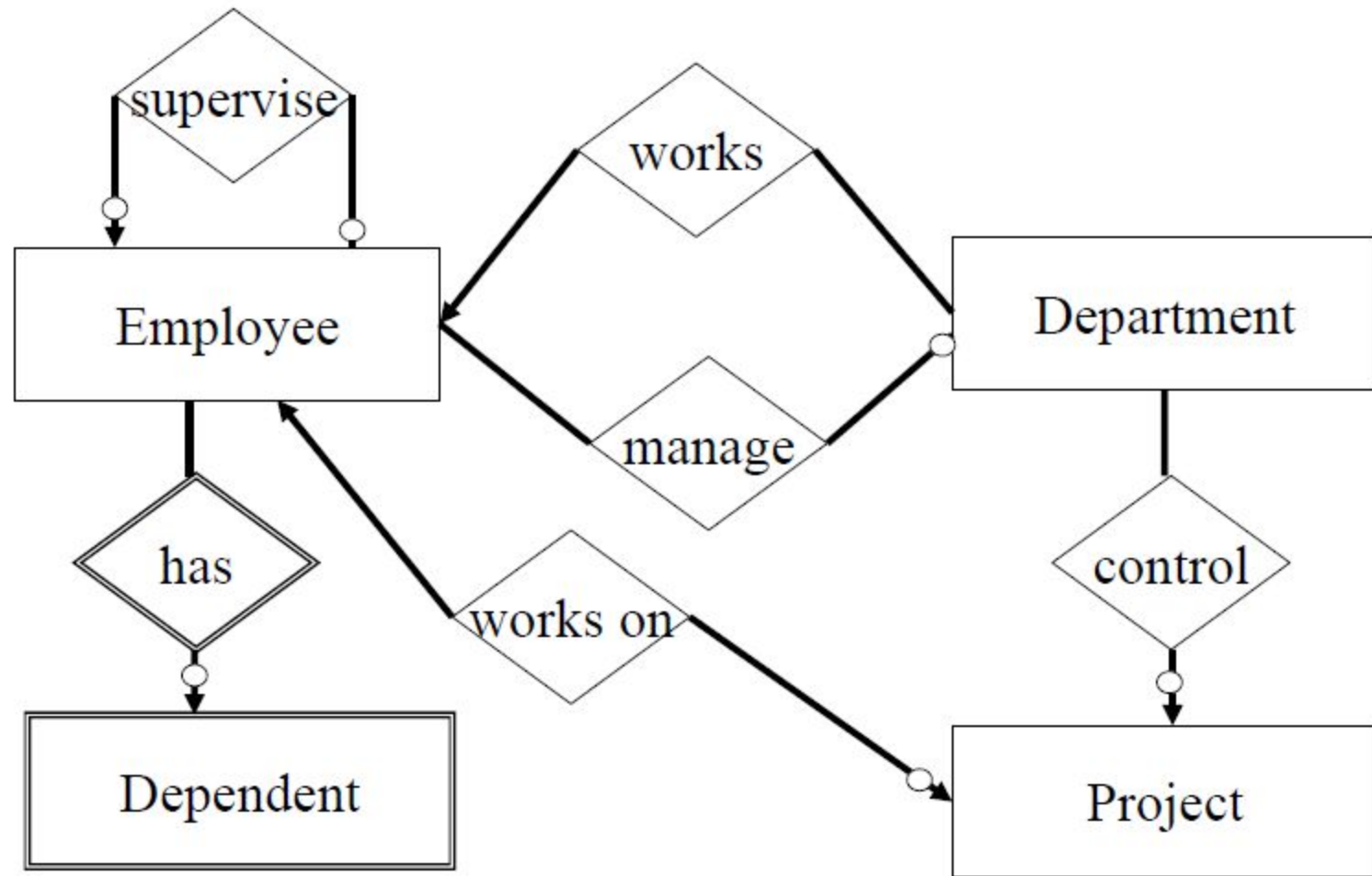
- A relationship between a weak entity type and its owner



has

A diamond shape with a double border, representing an identifying relationship.

Conceptual Design showing weak entities



Sample Entity Definitions

Name: **Department**

Type: Regular

Definition: a department of an organisation

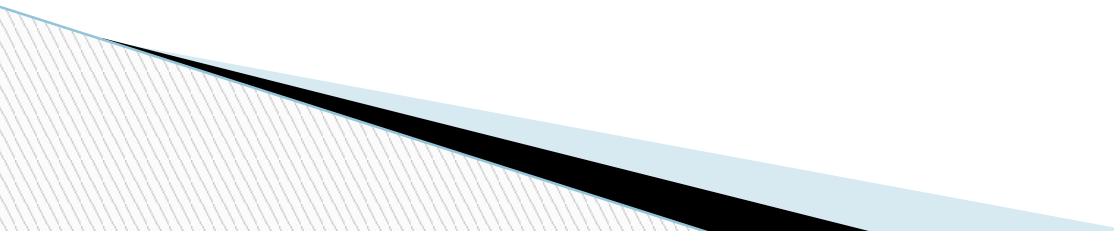
Identifier: Dept_No

Name: **Dependent**

Type: Weak

Definition: a person who is a dependent of an employee and entitle for insurance

Identifier: Depd_Name (partial only)



Sample Attribute Definitions

- Name: **Emp_No**

Domain: employee identities

Definition: unique identifier of an employee

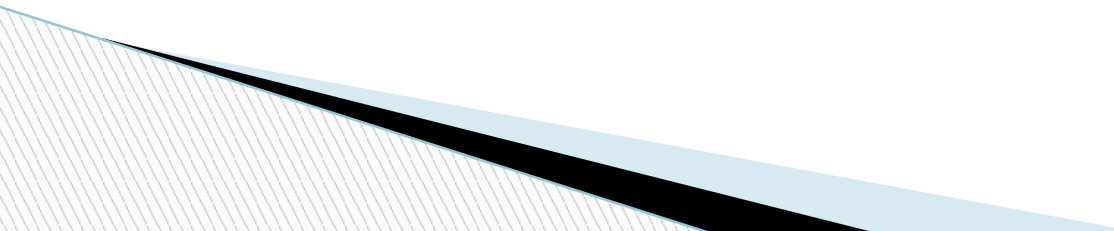
Null: No

- Name: **Emp_Name**

Components: First_Name, Mid_Initials,
Last_Name

Definition: a partial identifier of a name

Null: No



Sample Relationship Definitions

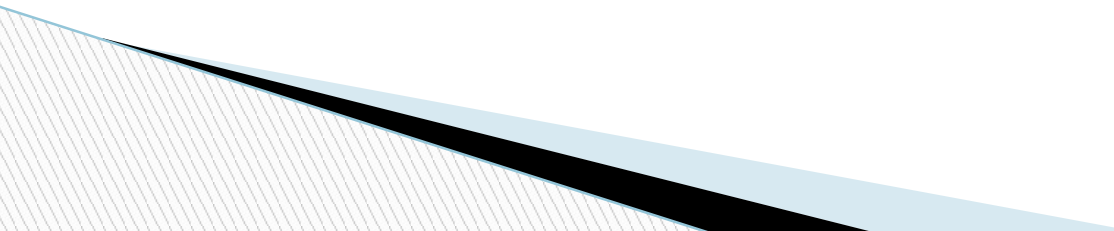
Name: **Works_for**

Type: binary 1:M

Definition: associates each employee with a department

Constraint: each employee must be attached to a department

Attributes: none



Domain Constraints

A specification of the characteristics of the data values that can be associated with one or more attributes

- Sample Domain Constraints
 - Name: **Employee identities**
 - Data Type: character
 - Length: 5
 - Allowable Characters: digits
- Name: **Last name**
 - Data Type: character
 - Max Length: 20
- Name: **DOB**
 - Data Type: date
 - Format: dd/mm/yyyy
 - dd = day
 - mm = month
 - yyyy = year

Associative Entity

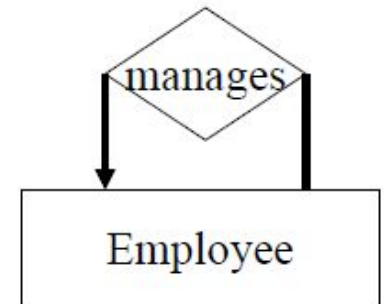
- ❑ An entity type that associates the instances of one or more entity types and contains attributes that are peculiar to the relationship between those entity instances
- ❑ All of the relationships for the participating entity types are “many” relationships
- ❑ The resulting associative entity type has independent meaning to end users, and preferably can be identified with a single-attribute identifier
- ❑ The associative entity has one or more attributes, in addition to the identifier
- ❑ The associative entity participates in one or more relationships independent of the entities related in the associated relationships



Relationships

□ Unary Relationship

- A relationship between the instances of a single entity type
- e.g. Person is married to a Person (1:1)
- Employee manages Employees (1:M)

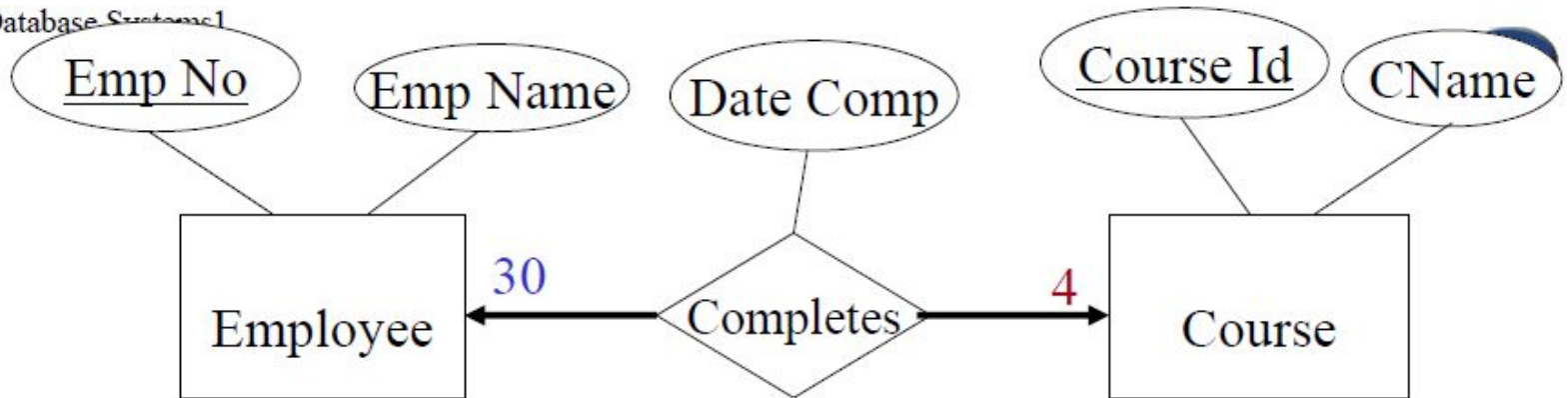


□ Binary Relationship

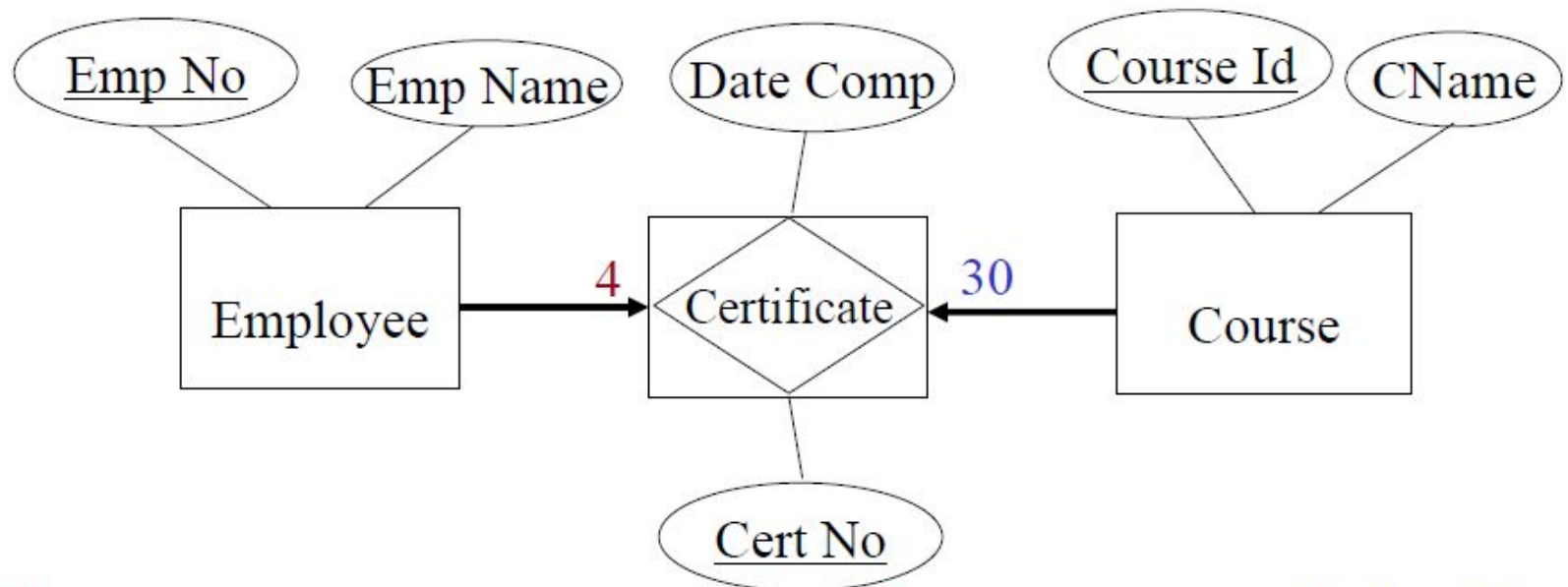
- A relationship between the instances of two entity types

□ Ternary Relationship

- A simultaneous relationship among the instances of three entity types
- can be treated as two many to many relationships

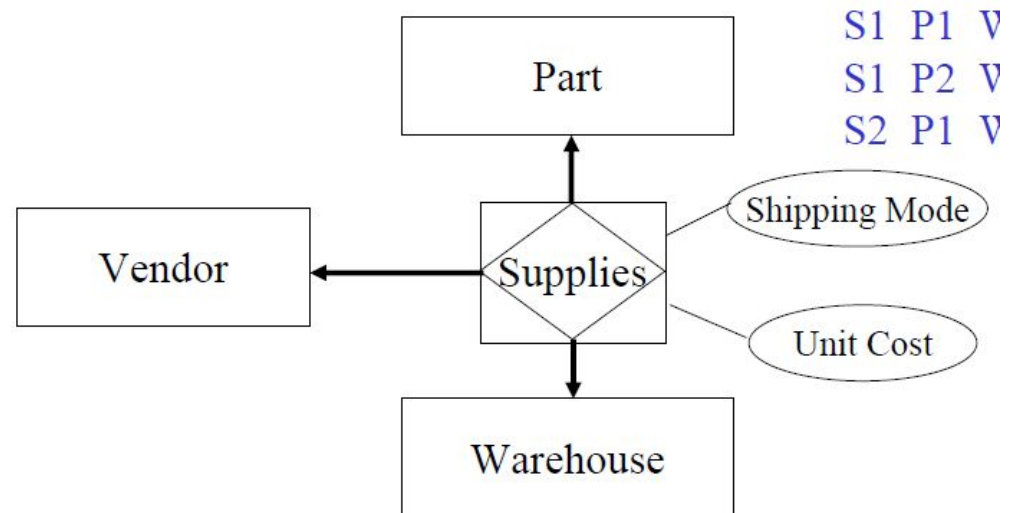
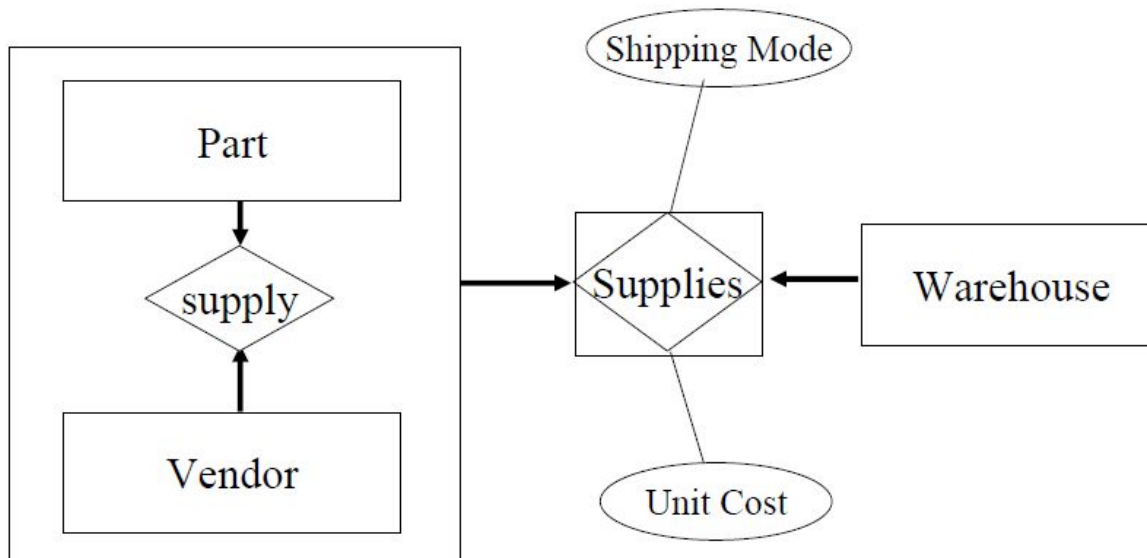


1 many to may relationship



2 one to many relationships

Ternary Relationship



Enhanced ERM

- Enhanced Entity-Relationship Model
 - The model that has resulted from extending the original E-R model with new modelling constructs
- Most important modelling construct incorporated is **Supertype / Subtype relationships**

Subtype

- A sub-grouping of the entities in an entity type that is meaningful to the organisation and that shares common attributes or relationships distinct from other subgrouping.
e.g. Student → Graduate, Undergraduate

Enhanced ERM

Supertype

- A generic entity type that has a relationship with one or more subtypes. e.g. Student

Attribute Inheritance

- A property that subtype entities inherit values of all attributes of the supertype

Generalisation

- The process of defining a more general entity type from a set of more specialised entity types

Specialisation

- The process of defining one or more subtypes of the supertype and forming supertype/subtype relationships
- 